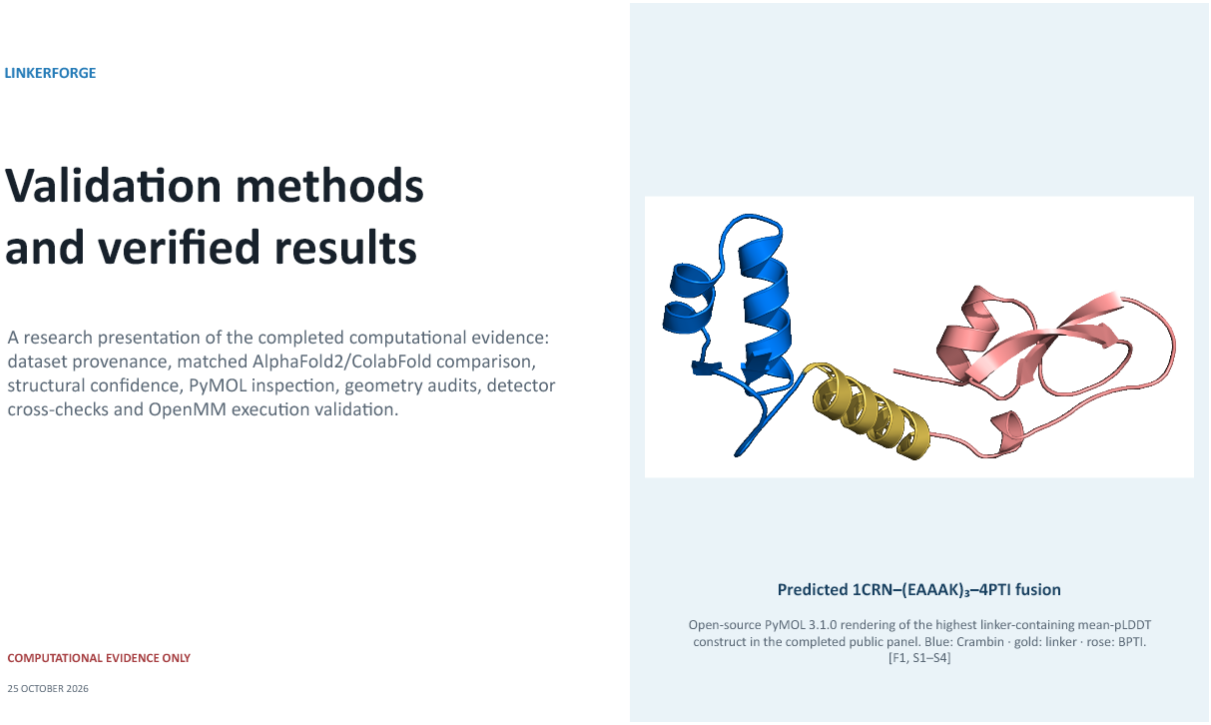
LinkerForge Presentation Study Guide and Speech

A complete guide to understanding, explaining and defending the validation methods and verified results presentation

Prepared 25 October 2026 | Computational evidence only



This document has two purposes. Part One teaches the project and the scientific meaning of every slide. Part Two provides a presentation speech that can be practised aloud. The final sections provide a terminology reference, difficult examiner questions and a claim-boundary checklist.

Main conclusion: LinkerForge currently demonstrates a reproducible, provenance-aware computational workflow for comparing fusion-linker candidates and inspecting structural evidence. It does not yet prove which linker is biologically optimal.

How to use this document

Do not try to memorise the entire document word for word. First study Sections 1 to 7 to understand the project. Then use the slide guide to connect each visual to the correct scientific interpretation. Finally practise the speech and use the question bank to test whether you can explain the work without overclaiming.

1. Learn the central research problem and why linker selection is context dependent.
2. Learn the difference between LinkerForge workflow evidence, AlphaFold confidence, geometry measurements, molecular dynamics and laboratory evidence.
3. Practise explaining every graph from its axes, units and individual marks before giving a conclusion.
4. Memorise the five prohibited overclaims in the claim-boundary checklist.
5. Practise the speech once with the slides and once without them.

Document map

- Part One: project and scientific study guide
- Part Two: slide-by-slide explanation of all 36 slides
- Part Three: complete presentation speech
- Part Four: difficult viva and examiner questions
- Part Five: rapid glossary and final claim-boundary checklist

Part One Project and scientific study guide

1 The research problem

Fusion proteins combine two protein regions into one translated amino-acid chain. The junction is not merely a typing detail. A linker can alter the distance between domains, their freedom of movement, their relative orientation, the accessibility of important residues, local steric packing and the ability of each domain to maintain a useful fold. A linker that works for one domain pair may fail for another because the geometry, surface chemistry, termini and functional constraints of the partners are different.

LinkerForge addresses this as a workflow problem. It accepts two protein-domain sequences, constructs complete fusion sequences with candidate and control linkers, passes those constructs to structure-prediction and analysis tools, preserves provenance and presents the resulting evidence in one interface. The current evidence validates execution, comparison and interpretation components. It does not validate a scientifically trained universal compatibility predictor.

2 What LinkerForge has actually completed

- A provenance-labelled inventory of 27,766 linker-related records with mined, natural and engineered evidence kept separate.
- A 971-record dual-curator review packet, family clustering and scientific preflight that correctly prevents unsupported ranker training.
- A matched public structural comparison using 3 domain pairs, 7 conditions per pair, 21 complete constructs and 105 retained AlphaFold2-pTM outputs.
- Extraction of overall pLDDT, linker pLDDT, pTM and cross-domain PAE.
- PyMOL and embedded-viewer inspection of saved PDB coordinates.
- C-alpha continuity, reference-domain RMSD, protected-site distance and accessibility, hydrogen-aware overlap and surface-burial measurements.
- An orthogonal DFLpred sensitivity check of flexible-linker character.
- Three 50 ps CUDA/OpenMM smoke replicates and automated software/schema verification.
- Versioned and provenance-bearing result artifacts, including source identities, software versions, configuration and file hashes.

3 The evidence ladder

Evidence level	What it establishes	What it does not establish
Software verification	The code, schemas and workflows execute as expected under the tested environment.	That a predicted linker works biologically.
Matched computational comparison	Candidate and controls received the same prediction settings and evaluation rules.	Generalisation to arbitrary proteins or a causal biological effect.
Predictor confidence	How confident AlphaFold2 is in local geometry or relative arrangement.	Expression, activity, thermodynamic stability or LinkerForge accuracy.
Structural descriptors	Properties of a saved predicted pose, such as RMSD, clashes, burial and accessibility.	A complete solution ensemble or retained function.
Short MD smoke test	The preparation, CUDA/OpenMM run and metric-extraction pipeline works.	Equilibrium, convergence or production stability.

Evidence level	What it establishes	What it does not establish
Outcome-linked validation	Would connect rankings to measured fusion outcomes.	Not present in the completed evidence.
Experimental validation	Would test expression, stability, activity or structure in the laboratory.	Outside the completed computational scope.

4 The planned model versus the completed workflow

A common examiner trap is to ask for the architecture as though it had already produced the presented ranking. The frozen project design proposes frozen ESM-2 protein embeddings, attachment-point geometric features processed by a graph neural network, a small linker encoder and multitask ranking heads. The intended learning task is pairwise or contrastive compatibility ranking, not a simple success/failure classifier. Candidate production for the MVP is constrained enumeration, mutation and retrieval; learned generation and diffusion remain future work.

The validated presentation does not report a trained or calibrated version of that ranker. The highlighted EAAAK construct was selected from a matched AlphaFold2/ColabFold comparison. It is therefore incorrect to call 93.8 a LinkerForge prediction accuracy or to say that the proposed ranker selected the experimentally best linker.

5 Why the dataset cannot yet train the scientific ranker

The 27,766 records mainly answer an identity question: where might a linker occur, or what linker sequence has been described? The desired compatibility ranker requires an outcome question: when a particular linker was inserted between particular partners, what happened to expression, solubility, activity, stability, structure and other relevant outcomes? Natural interdomain linkers are useful for representation learning, but a native multidomain protein has evolved in a different context from an engineered fusion.

Data type	Available evidence	Use	Limit
Mined candidates	Sequence-model or rule-based candidate spans	Prioritisation and curation	Not ground-truth compatibility
Natural CATH/PDB spans	Domain boundaries supported by public structures	Natural reference and possible pretraining	Not an engineered-fusion outcome
DisProt DFL records	Curated disorder/flexible-linker annotations	Flexible-linker identity reference	Not universal fusion success
Published templates	Documented linker sequences	Controls and retrieval	Published use is context specific
Required outcome data	Partner-linker-partner construct plus measured outcomes	Train and calibrate compatibility ranking	Currently insufficient

6 Why the benchmark is protected and fail closed

A prospective benchmark must remain independent of model development. Predictions, feature choices, thresholds and analysis rules are fixed before the protected outcomes are revealed. If protected labels, public scores or previously revealed outcomes are used to select the model, the final result becomes optimistic because the test set has indirectly become part of development.

Fail closed means that the workflow refuses to continue when the independence conditions are not satisfied. This is a scientific safeguard. It is not the same as a software failure, and it should not be bypassed merely to produce a favorable number.

7 How to interpret the highest-confidence construct

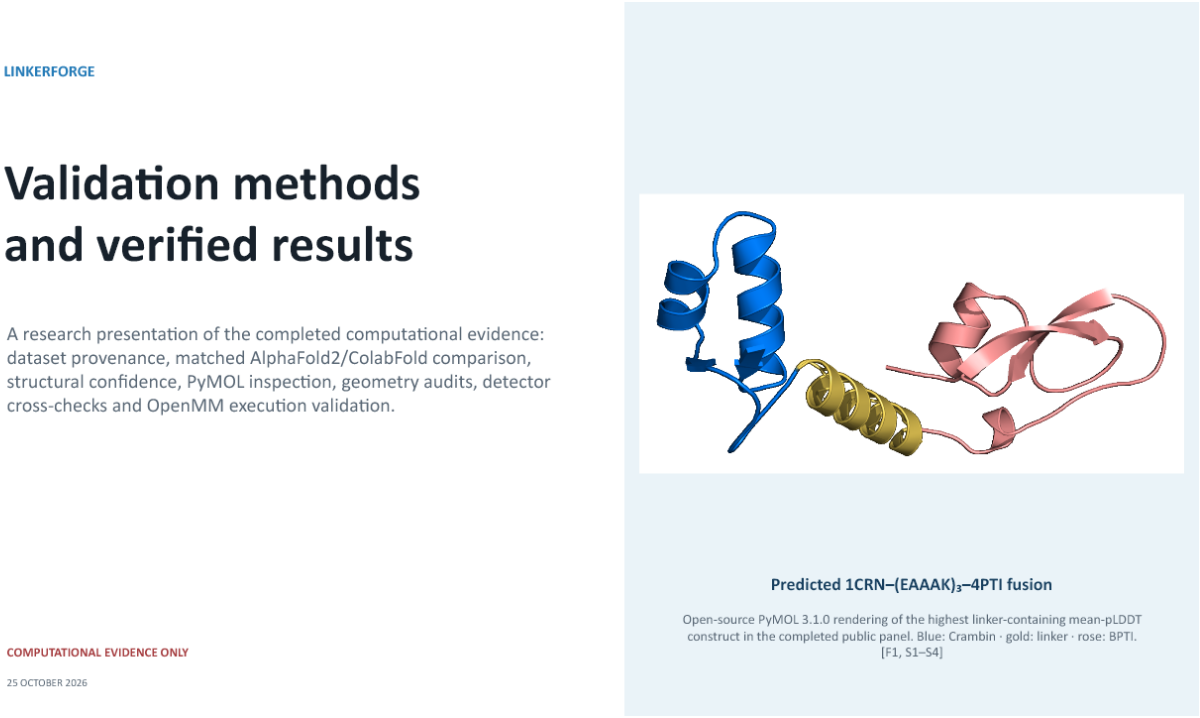
The construct 1CRN-EAAAK15-4PTI had the highest mean pLDDT among the 18 linker-containing constructs in the completed 21-construct public panel. Its overall mean pLDDT was 93.8, linker mean pLDDT 84.9, pTM 0.69 and mean cross-domain PAE 9.7 A. This is a comparatively favorable AlphaFold2 confidence profile for that domain pair and protocol.

It is not the highest overall row if zero-linker controls are included: the 1PGA-1LMB direct-fusion control reached 94.9 mean pLDDT. The selection of the EAAAK example was post hoc and descriptive, so it is suitable for demonstrating the application but not for claiming prospective superiority.

Part Two Slide by slide explanation

Each entry contains the purpose of the slide, the scientific meaning of its content, the correct interpretation and the transition to the next idea. The slide image is included only as a navigation aid; the source PowerPoint has not been modified.

Slide 1 Validation methods and verified results



Slide 1 from the unchanged LinkerForge validation presentation

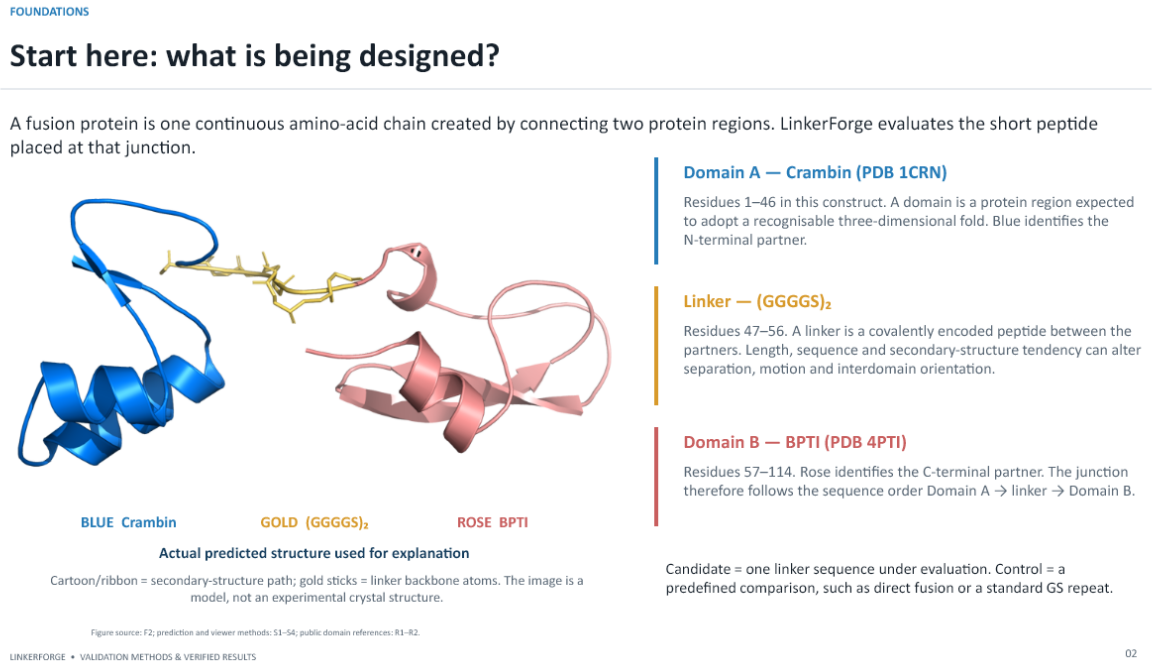
Purpose: Establish the scope: this is a report of completed computational evidence, not an experimental validation claim.

What to understand: The cover structure is the predicted 1CRN-EAAAK15-4PTI fusion.

- Blue is Crambin, gold is the linker and rose is BPTI.
- PyMOL rendered saved coordinates; it did not predict or simulate them.

How to present it: Open by saying what the workflow proves and what it does not prove.

Slide 2 What is being designed



Slide 2 from the unchanged LinkerForge validation presentation

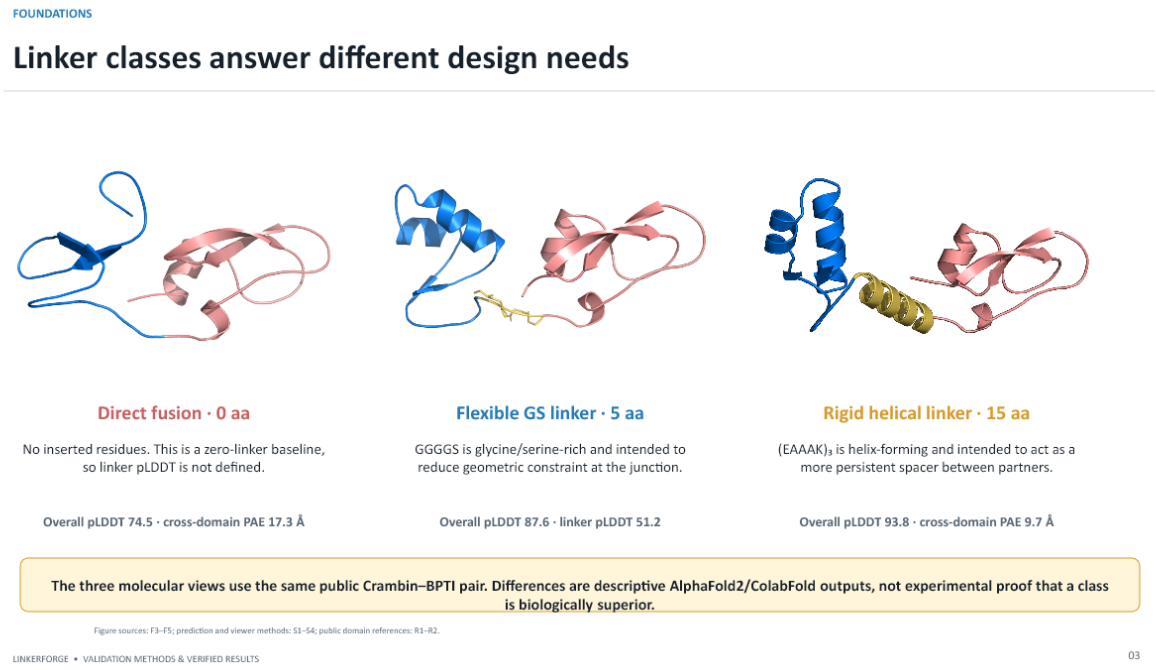
Purpose: Teach the audience the fusion sequence before discussing metrics.

What to understand: A fusion is one continuous amino-acid chain.

- Domain A occupies the N-terminal side, the linker is the junction, and Domain B occupies the C-terminal side.
- The shown GS10 structure is a computational model. A ribbon shows the backbone path; sticks show selected atoms.

How to present it: Transition from the biological object to the design choices at its junction.

Slide 3 Linker classes



Slide 3 from the unchanged LinkerForge validation presentation

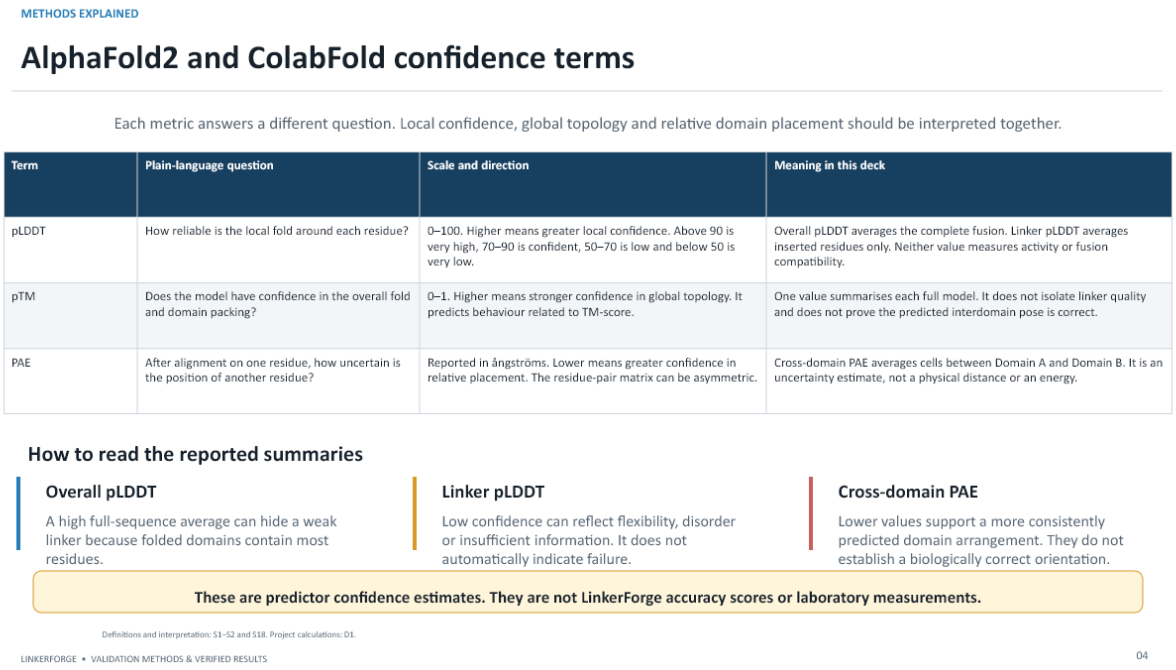
Purpose: Explain why direct, flexible and rigid linkers answer different design questions.

What to understand: Direct fusion is the zero-linker baseline.

- GGGGS is GS-rich and intended to allow motion.
- EAAAK repeats are intended to form a more persistent helical spacer.
- The displayed numerical differences are descriptive for Crambin-BPTI only.

How to present it: Emphasise that linker class is a design hypothesis, not a universal ranking.

Slide 4 AlphaFold2 and ColabFold confidence



METHODS EXPLAINED

AlphaFold2 and ColabFold confidence terms

Each metric answers a different question. Local confidence, global topology and relative domain placement should be interpreted together.

Term	Plain-language question	Scale and direction	Meaning in this deck
pLDDT	How reliable is the local fold around each residue?	0–100. Higher means greater local confidence. Above 90 is very high, 70–90 is confident, 50–70 is low and below 50 is very low.	Overall pLDDT averages the complete fusion. Linker pLDDT averages inserted residues only. Neither value measures activity or fusion compatibility.
pTM	Does the model have confidence in the overall fold and domain packing?	0–1. Higher means stronger confidence in global topology. It predicts behaviour related to TM-score.	One value summarises each full model. It does not isolate linker quality and does not prove the predicted interdomain pose is correct.
PAE	After alignment on one residue, how uncertain is the position of another residue?	Reported in ångströms. Lower means greater confidence in relative placement. The residue-pair matrix can be asymmetric.	Cross-domain PAE averages cells between Domain A and Domain B. It is an uncertainty estimate, not a physical distance or an energy.

How to read the reported summaries

Overall pLDDT

A high full-sequence average can hide a weak linker because folded domains contain most residues.

Linker pLDDT

Low confidence can reflect flexibility, disorder or insufficient information. It does not automatically indicate failure.

Cross-domain PAE

Lower values support a more consistently predicted domain arrangement. They do not establish a biologically correct orientation.

These are predictor confidence estimates. They are not LinkerForge accuracy scores or laboratory measurements.

Definitions and interpretation: S1–S2 and S18. Project calculations: D1.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 04

Slide 4 from the unchanged LinkerForge validation presentation

Purpose: Separate local confidence, global topology and interdomain uncertainty.

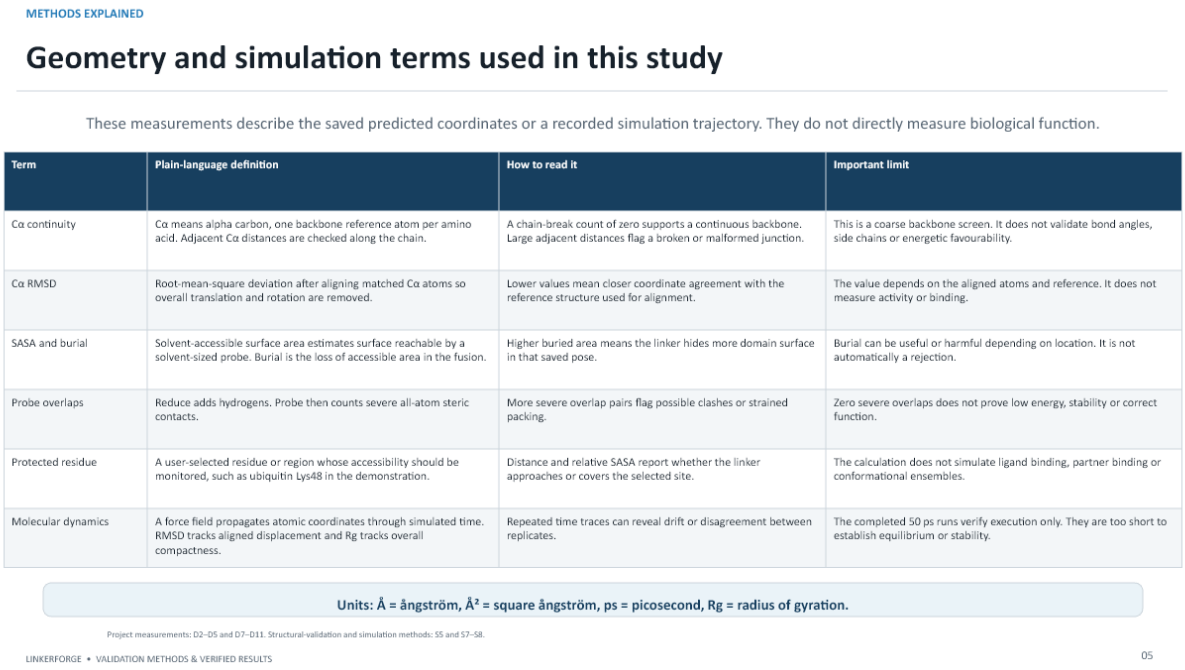
What to understand: pLDDT is per-residue local confidence on a 0-100 scale.

- pTM is global topology confidence on a 0-1 scale.
- PAE estimates positional uncertainty after alignment and is reported in A.
- A high average pLDDT can hide a low-confidence linker.

How to present it: Say explicitly that none of these values is LinkerForge accuracy.

Do not overclaim: Never convert pLDDT into percent accuracy.

Slide 5 Geometry and simulation terms



METHODS EXPLAINED

Geometry and simulation terms used in this study

These measurements describe the saved predicted coordinates or a recorded simulation trajectory. They do not directly measure biological function.

Term	Plain-language definition	How to read it	Important limit
Cα continuity	Cα means alpha carbon, one backbone reference atom per amino acid. Adjacent Cα distances are checked along the chain.	A chain-break count of zero supports a continuous backbone. Large adjacent distances flag a broken or malformed junction.	This is a coarse backbone screen. It does not validate bond angles, side chains or energetic favourability.
Cα RMSD	Root-mean-square deviation after aligning matched Cα atoms so overall translation and rotation are removed.	Lower values mean closer coordinate agreement with the reference structure used for alignment.	The value depends on the aligned atoms and reference. It does not measure activity or binding.
SASA and burial	Solvent-accessible surface area estimates surface reachable by a solvent-sized probe. Burial is the loss of accessible area in the fusion.	Higher buried area means the linker hides more domain surface in that saved pose.	Burial can be useful or harmful depending on location. It is not automatically a rejection.
Probe overlaps	Reduce adds hydrogens. Probe then counts severe all-atom steric contacts.	More severe overlap pairs flag possible clashes or strained packing.	Zero severe overlaps does not prove low energy, stability or correct function.
Protected residue	A user-selected residue or region whose accessibility should be monitored, such as ubiquitin Lys48 in the demonstration.	Distance and relative SASA report whether the linker approaches or covers the selected site.	The calculation does not simulate ligand binding, partner binding or conformational ensembles.
Molecular dynamics	A force field propagates atomic coordinates through simulated time. RMSD tracks aligned displacement and Rg tracks overall compactness.	Repeated time traces can reveal drift or disagreement between replicates.	The completed 50 ps runs verify execution only. They are too short to establish equilibrium or stability.

Units: Å = ångström, Å² = square ångström, ps = picosecond, Rg = radius of gyration.

Project measurements: D2–D5 and D7–D11. Structural-validation and simulation methods: S5 and S7–S8.

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05

Slide 5 from the unchanged LinkerForge validation presentation

Purpose: Define every structural descriptor before it appears in the results.

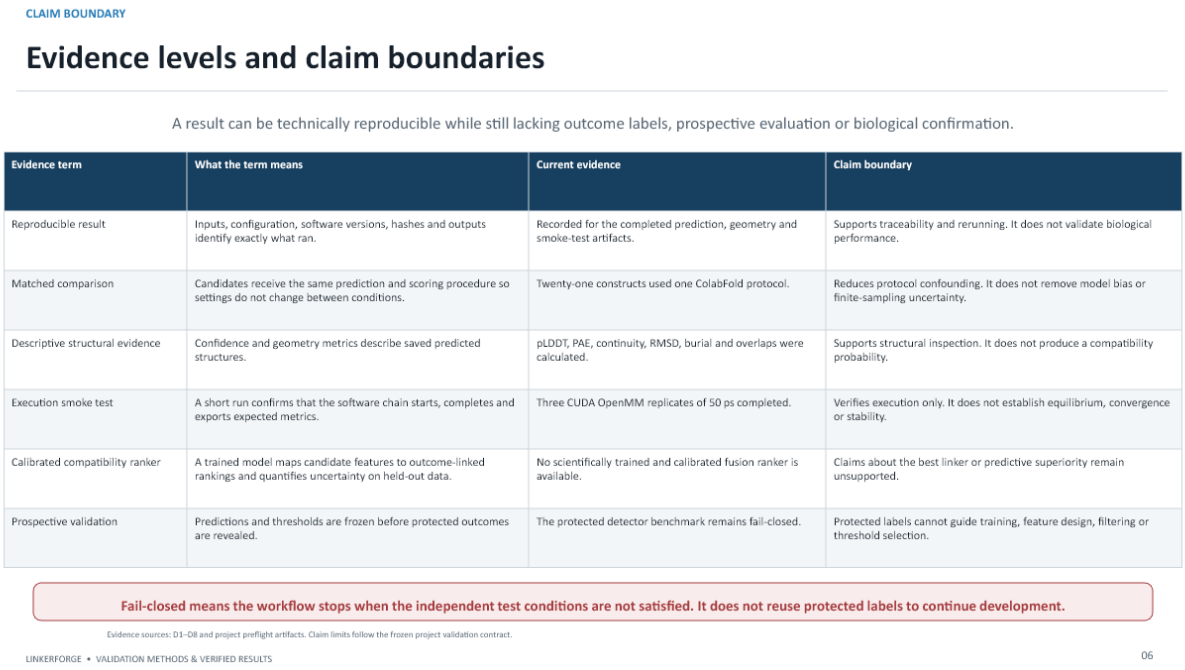
What to understand: C-alpha continuity screens for coordinate gaps.

- C-alpha RMSD measures reference-aligned coordinate difference.
- SASA and burial describe solvent exposure.
- Reduce and Probe provide hydrogen-aware contact assessment.
- OpenMM trajectories provide time-dependent coordinates; Rg describes compactness.

How to present it: Use this slide as the vocabulary foundation for slides 21-26.

Do not overclaim: A geometric descriptor is not a biological assay.

Slide 6 Evidence levels and claim boundaries



CLAIM BOUNDARY

Evidence levels and claim boundaries

A result can be technically reproducible while still lacking outcome labels, prospective evaluation or biological confirmation.

Evidence term	What the term means	Current evidence	Claim boundary
Reproducible result	Inputs, configuration, software versions, hashes and outputs identify exactly what ran.	Recorded for the completed prediction, geometry and smoke-test artifacts.	Supports traceability and rerunning. It does not validate biological performance.
Matched comparison	Candidates receive the same prediction and scoring procedure so settings do not change between conditions.	Twenty-one constructs used one ColabFold protocol.	Reduces protocol confounding. It does not remove model bias or finite-sampling uncertainty.
Descriptive structural evidence	Confidence and geometry metrics describe saved predicted structures.	pLDDT, PAE, continuity, RMSD, burial and overlaps were calculated.	Supports structural inspection. It does not produce a compatibility probability.
Execution smoke test	A short run confirms that the software chain starts, completes and exports expected metrics.	Three CUDA OpenMM replicates of 50 ps completed.	Verifies execution only. It does not establish equilibrium, convergence or stability.
Calibrated compatibility ranker	A trained model maps candidate features to outcome-linked rankings and quantifies uncertainty on held-out data.	No scientifically trained and calibrated fusion ranker is available.	Claims about the best linker or predictive superiority remain unsupported.
Prospective validation	Predictions and thresholds are frozen before protected outcomes are revealed.	The protected detector benchmark remains fail-closed.	Protected labels cannot guide training, feature design, filtering or threshold selection.

Fail-closed means the workflow stops when the independent test conditions are not satisfied. It does not reuse protected labels to continue development.

Evidence sources: D1–D8 and project preflight artifacts. Claim limits follow the frozen project validation contract.

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Slide 6 from the unchanged LinkerForge validation presentation

Purpose: Show that reproducibility, confidence, structural description and biological proof are different evidence levels.

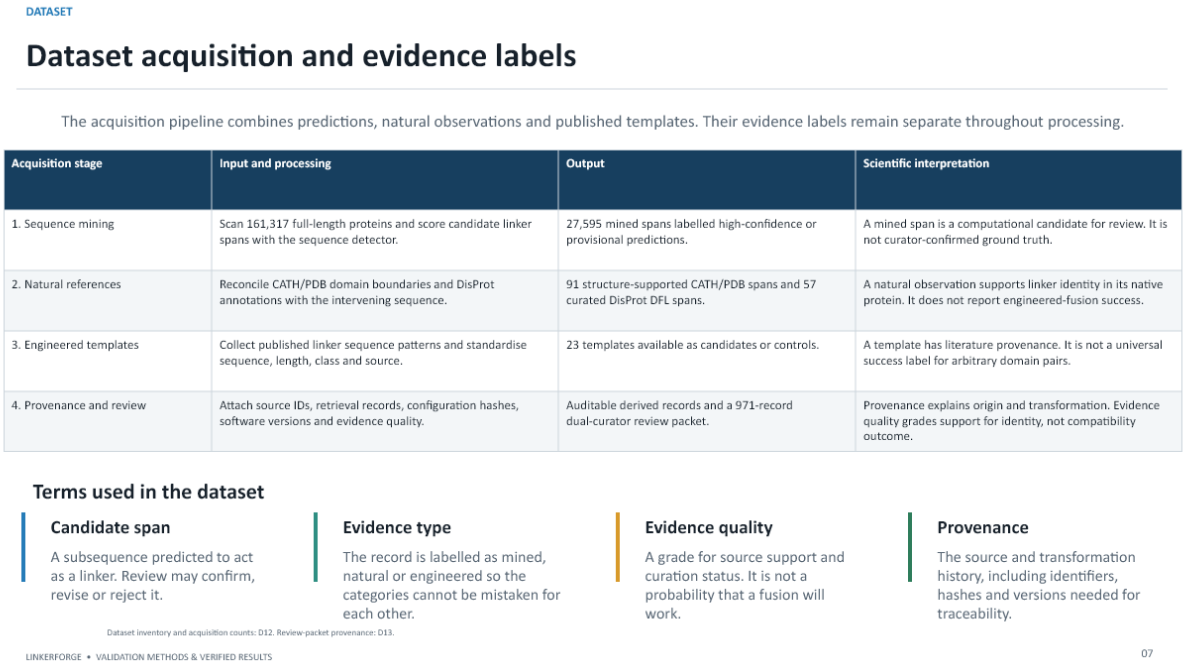
What to understand: A reproducible run can still lack outcome labels.

- A matched comparison reduces protocol confounding but does not remove model bias.
- A 50 ps smoke test validates execution, not stability.
- A calibrated ranker and prospective validation are not yet available.

How to present it: Explain fail-closed behaviour as protection against data leakage.

Do not overclaim: Technical reproducibility and scientific validation are not synonyms.

Slide 7 Dataset acquisition



DATASET

Dataset acquisition and evidence labels

The acquisition pipeline combines predictions, natural observations and published templates. Their evidence labels remain separate throughout processing.

Acquisition stage	Input and processing	Output	Scientific interpretation
1. Sequence mining	Scan 161,317 full-length proteins and score candidate linker spans with the sequence detector.	27,595 mined spans labelled high-confidence or provisional predictions.	A mined span is a computational candidate for review. It is not curator-confirmed ground truth.
2. Natural references	Reconcile CATH/PDB domain boundaries and DisProt annotations with the intervening sequence.	91 structure-supported CATH/PDB spans and 57 curated DisProt DFL spans.	A natural observation supports linker identity in its native protein. It does not report engineered-fusion success.
3. Engineered templates	Collect published linker sequence patterns and standardise sequence, length, class and source.	23 templates available as candidates or controls.	A template has literature provenance. It is not a universal success label for arbitrary domain pairs.
4. Provenance and review	Attach source IDs, retrieval records, configuration hashes, software versions and evidence quality.	Auditable derived records and a 971-record dual-curator review packet.	Provenance explains origin and transformation. Evidence quality grades support for identity, not compatibility outcome.

Terms used in the dataset

Candidate span
A subsequence predicted to act as a linker. Review may confirm, revise or reject it.

Evidence type
The record is labelled as mined, natural or engineered so the categories cannot be mistaken for each other.

Evidence quality
A grade for source support and curation status. It is not a probability that a fusion will work.

Provenance
The source and transformation history, including identifiers, hashes and versions needed for traceability.

Dataset inventory and acquisition counts: D12. Review-packet provenance: D13.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS

07

Slide 7 from the unchanged LinkerForge validation presentation

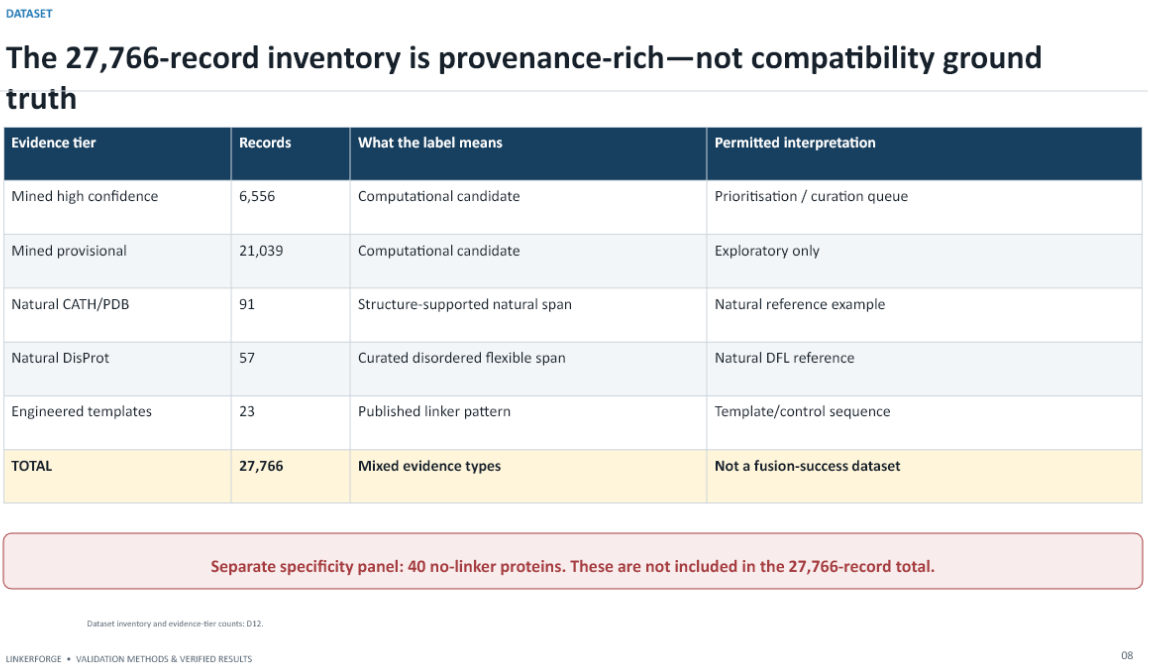
Purpose: Explain how mined candidates, natural references and engineered templates were acquired and kept separately labelled.

What to understand: 161,317 proteins were scanned.

- 27,595 mined spans were identified.
- 91 CATH/PDB and 57 DisProt natural references were collected.
- 23 engineered templates were recorded.
- Provenance includes identifiers, transformations, hashes and versions.

How to present it: Stress that evidence quality is not a probability of fusion success.

Slide 8 Dataset composition



DATASET

The 27,766-record inventory is provenance-rich—not compatibility ground truth

Evidence tier	Records	What the label means	Permitted interpretation
Mined high confidence	6,556	Computational candidate	Prioritisation / curation queue
Mined provisional	21,039	Computational candidate	Exploratory only
Natural CATH/PDB	91	Structure-supported natural span	Natural reference example
Natural DisProt	57	Curated disordered flexible span	Natural DFL reference
Engineered templates	23	Published linker pattern	Template/control sequence
TOTAL	**27,766**	**Mixed evidence types**	**Not a fusion-success dataset**

Separate specificity panel: 40 no-linker proteins. These are not included in the 27,766-record total.

Dataset inventory and evidence-tier counts: D12.

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Slide 8 from the unchanged LinkerForge validation presentation

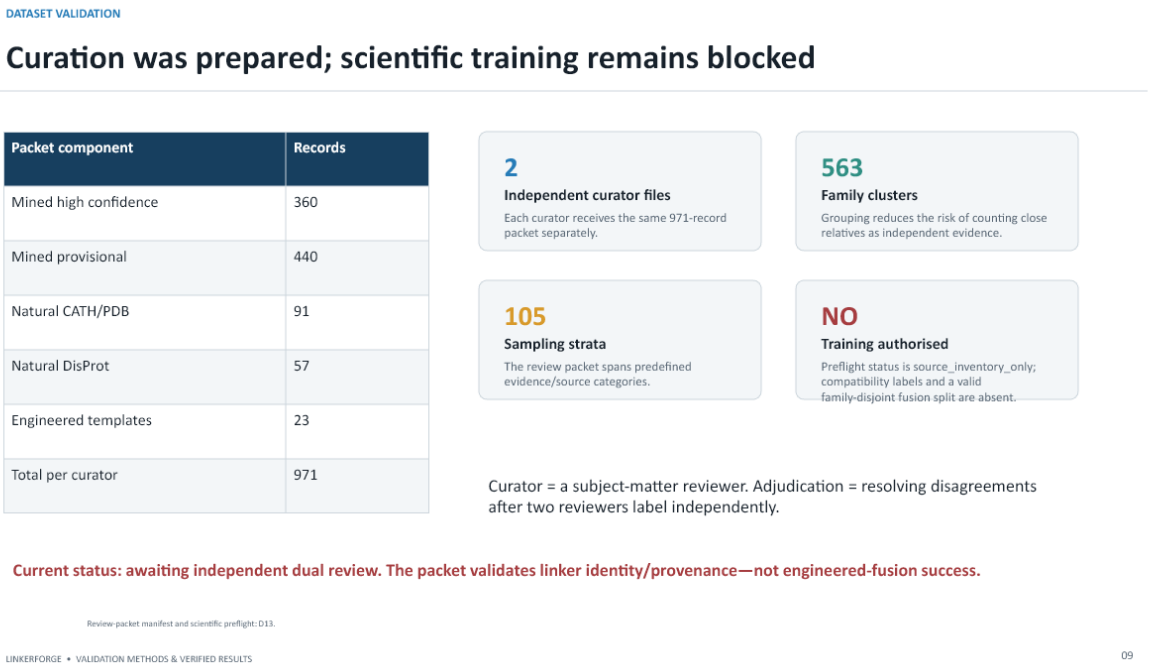
Purpose: Explain the exact 27,766 total and why it is not compatibility ground truth.

What to understand: 6,556 mined high-confidence and 21,039 provisional candidates dominate the inventory.

- 91 CATH/PDB, 57 DisProt and 23 engineered-template records provide stronger source context.
- A separate 40-protein no-linker panel is not part of the total.

How to present it: The inventory supports curation and pretraining research, not a best-linker claim.

Slide 9 Curation and training preflight



Slide 9 from the unchanged LinkerForge validation presentation

Purpose: Explain what the 971-record packet can validate and why training is blocked.

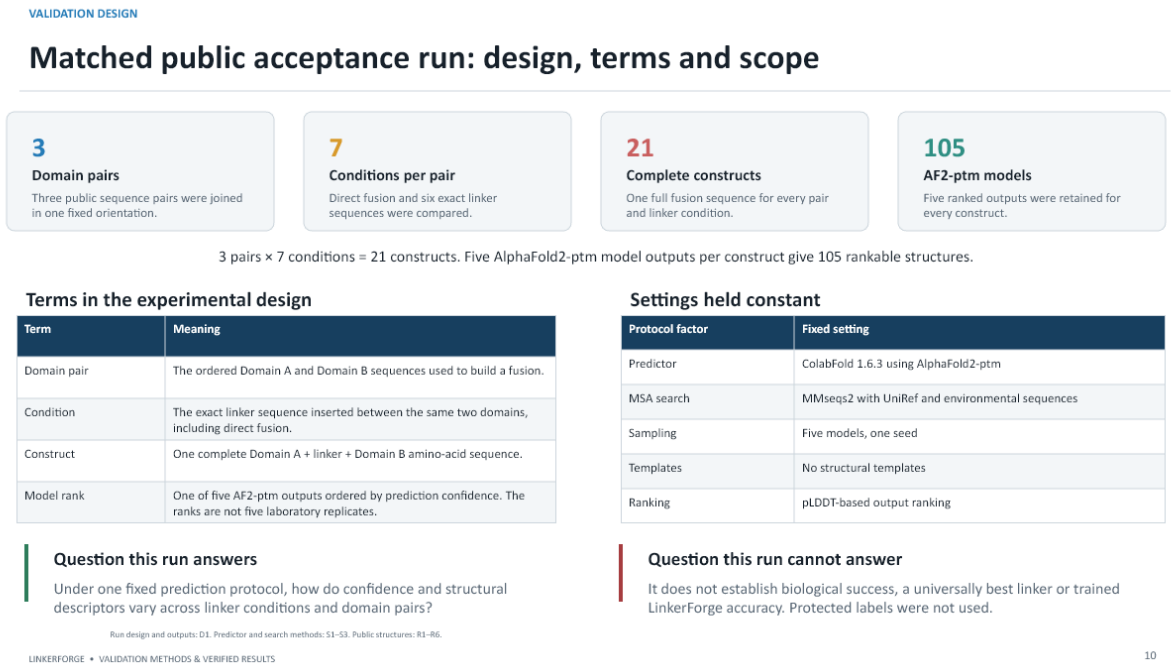
What to understand: Each of two curators receives the same packet independently.

- The packet spans 563 family clusters and 105 sampling strata.
- Adjudication resolves disagreements only after independent review.
- Compatibility labels and a valid family-disjoint fusion split are missing.

How to present it: Say that the block is scientifically correct rather than an implementation failure.

Do not overclaim: Curation validates identity and provenance, not engineered-fusion success.

Slide 10 Matched public acceptance run



Slide 10 from the unchanged LinkerForge validation presentation

Purpose: Define the experimental unit and the fixed protocol.

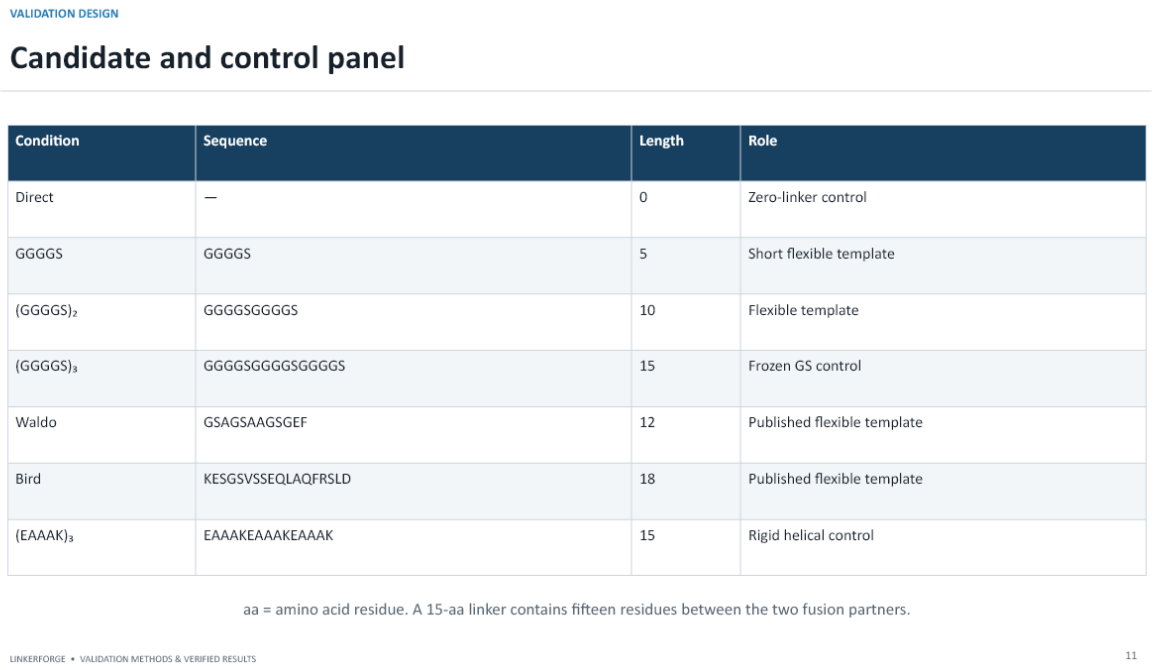
What to understand: 3 domain pairs x 7 conditions gives 21 complete constructs.

- Five AF2-pTM ranked outputs gives 105 structures.
- Five ranked models are not laboratory replicates.
- All candidates used the same ColabFold settings, one seed and no templates.

How to present it: The run measures prediction sensitivity to linker and partner context.

Do not overclaim: The five model ranks are correlated predictions, not independent experiments.

Slide 11 Candidate and control panel



VALIDATION DESIGN

Candidate and control panel

Condition	Sequence	Length	Role
Direct	—	0	Zero-linker control
GGGGS	GGGGS	5	Short flexible template
$(GGGGS)_2$	GGGGSGGGGS	10	Flexible template
$(GGGGS)_3$	GGGGSGGGGSGGGGS	15	Frozen GS control
Waldo	GSAGSAAGSGEF	12	Published flexible template
Bird	KESGSVSSEQLAQFRSLD	18	Published flexible template
$(EAAAK)_3$	EAAAKEAAAKEAAAK	15	Rigid helical control

aa = amino acid residue. A 15-aa linker contains fifteen residues between the two fusion partners.

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Slide 11 from the unchanged LinkerForge validation presentation

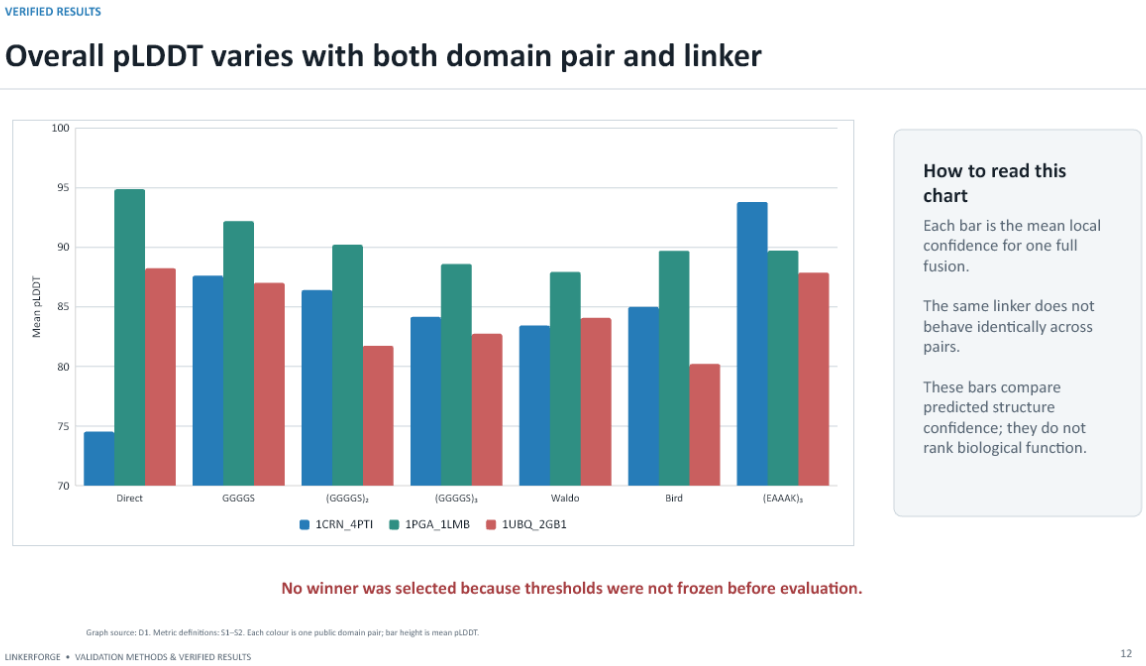
Purpose: Identify the seven exact conditions.

What to understand: Direct fusion is the zero-linker control.

- GGGGS, GS10 and GS15 provide a flexible length series.
- Waldo and Bird are published sequences.
- EAAAK15 is the rigid helical control.
- The fixed GS15 sequence is the project control specified in the design contract.

How to present it: Explain why controls are essential for interpreting a candidate.

Slide 12 Overall pLDDT across pairs



Slide 12 from the unchanged LinkerForge validation presentation

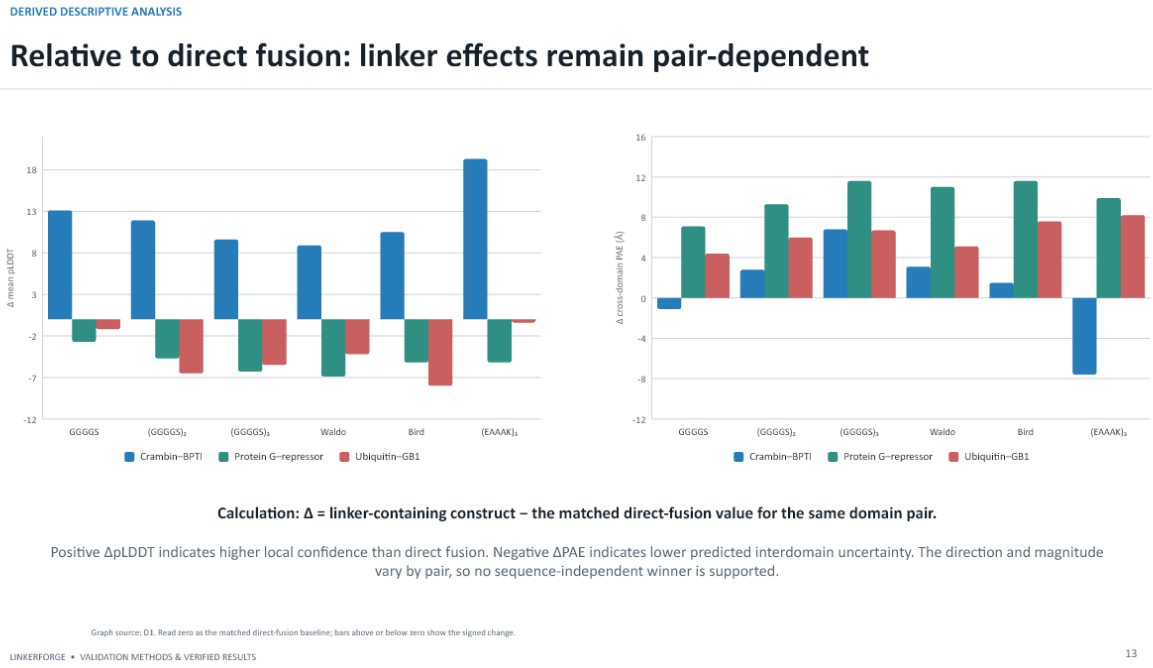
Purpose: Read one bar as one full fusion and compare within the same pair first.

What to understand: Crambin-BPTI showed its highest linker-containing pLDDT with EAAAK.

- Protein G-repressor was highest with direct fusion; short GGGGS led the linked conditions.
- Ubiquitin-GB1 showed direct and EAAAK close in overall pLDDT.
- The same linker behaved differently across pairs.

How to present it: The conclusion is partner dependence, not a universal winner.

Slide 13 Changes relative to direct fusion



Slide 13 from the unchanged LinkerForge validation presentation

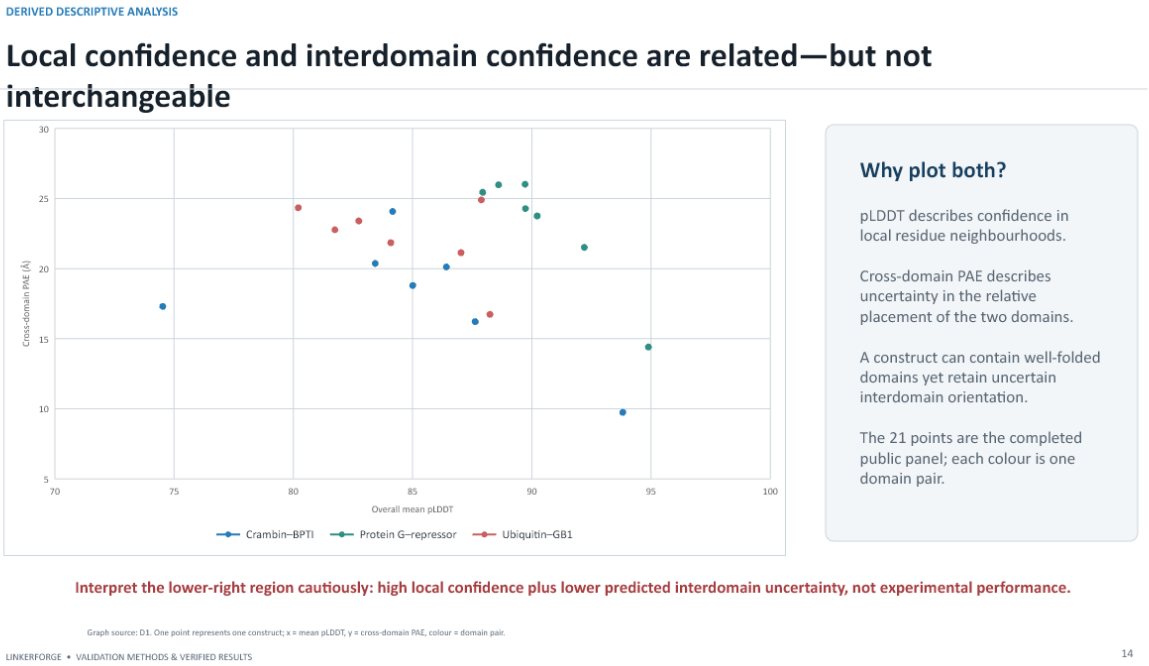
Purpose: Explain the signed delta calculation against the matched zero-linker baseline.

What to understand: Positive delta pLDDT means greater local confidence than direct fusion.

- Negative delta PAE means lower predicted interdomain uncertainty.
- Crambin-BPTI EAAAK improved both descriptors, while other pairs often favored direct fusion.

How to present it: A zero-centered comparison makes direction visible but does not turn confidence into function.

Slide 14 pLDDT versus cross-domain PAE



Slide 14 from the unchanged LinkerForge validation presentation

Purpose: Explain why local quality and interdomain placement must be evaluated together.

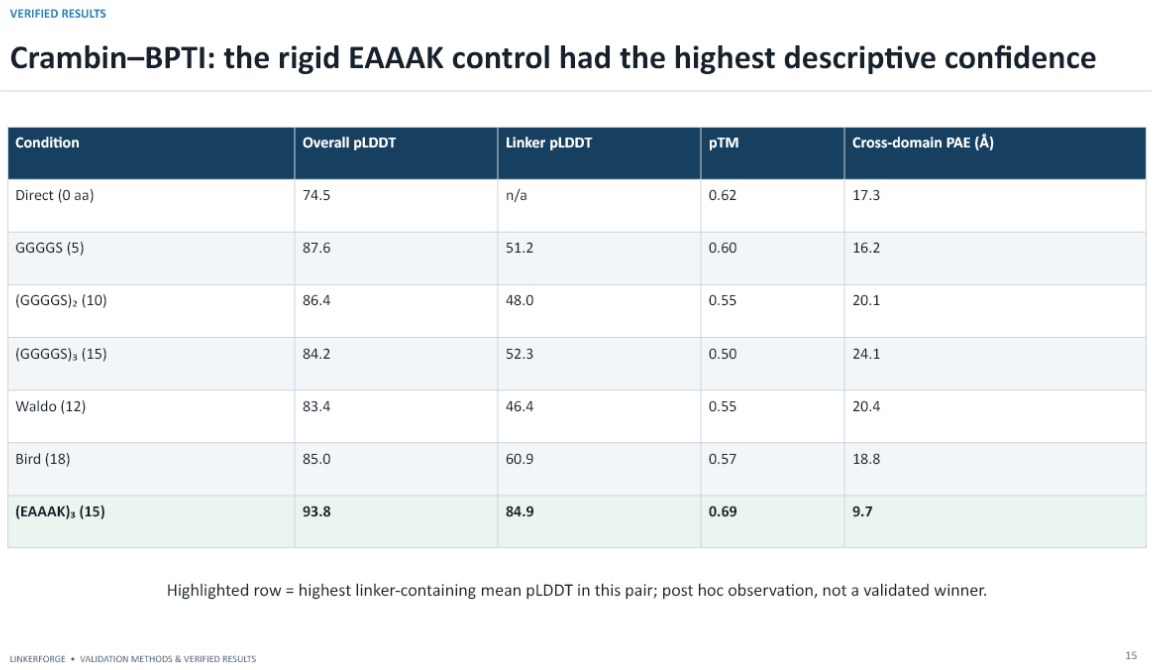
What to understand: Each point is one construct.

- Moving right means higher mean pLDDT.
- Moving down means lower cross-domain uncertainty.
- The lower-right region is descriptively favorable, not proof of performance.

How to present it: A structure can contain confident domains but an uncertain relative orientation.

Do not overclaim: The lower-right region is favorable only in predictor-confidence terms.

Slide 15 Crambin-BPTI table



VERIFIED RESULTS

Crambin–BPTI: the rigid EAAAK control had the highest descriptive confidence

Condition	Overall pLDDT	Linker pLDDT	pTM	Cross-domain PAE (Å)
Direct (0 aa)	74.5	n/a	0.62	17.3
GGGGS (5)	87.6	51.2	0.60	16.2
$(GGGGS)_2$ (10)	86.4	48.0	0.55	20.1
$(GGGGS)_3$ (15)	84.2	52.3	0.50	24.1
Waldo (12)	83.4	46.4	0.55	20.4
Bird (18)	85.0	60.9	0.57	18.8
$(EAAAK)_3$ (15)	**93.8**	**84.9**	**0.69**	**9.7**

Highlighted row = highest linker-containing mean pLDDT in this pair; post hoc observation, not a validated winner.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 15

Slide 15 from the unchanged LinkerForge validation presentation

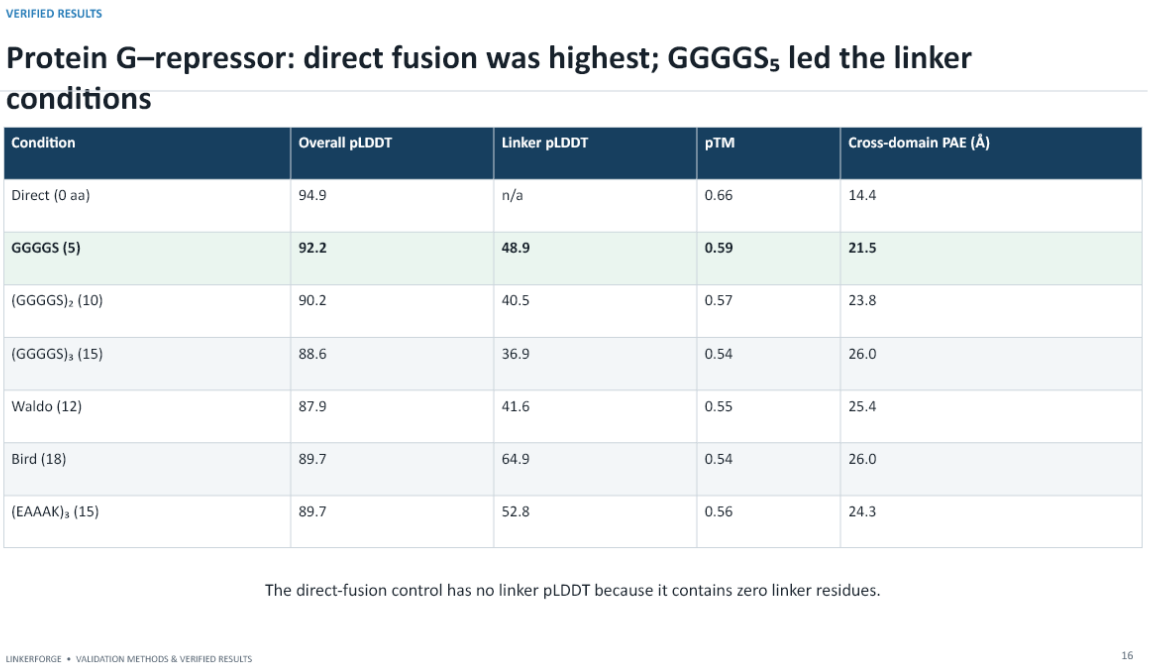
Purpose: Present the complete per-condition confidence profile and the post hoc highlight.

What to understand: EAAAK15: pLDDT 93.8, linker pLDDT 84.9, pTM 0.69, cross-domain PAE 9.7 A.

- Direct fusion was substantially lower in overall pLDDT at 74.5.
- GS linkers had lower linker-specific confidence, consistent with greater flexibility or uncertainty.

How to present it: Call EAAAK the highest linker-containing descriptive result for this pair, not a validated winner.

Slide 16 Protein G-repressor table



VERIFIED RESULTS

Protein G–repressor: direct fusion was highest; $GGGGS_5$ led the linker conditions

Condition	Overall pLDDT	Linker pLDDT	pTM	Cross-domain PAE (Å)
Direct (0 aa)	94.9	n/a	0.66	14.4
GGGGS (5)	**92.2**	**48.9**	**0.59**	**21.5**
$(GGGGS)_2$ (10)	90.2	40.5	0.57	23.8
$(GGGGS)_3$ (15)	88.6	36.9	0.54	26.0
Waldo (12)	87.9	41.6	0.55	25.4
Bird (18)	89.7	64.9	0.54	26.0
$(EAAAK)_3$ (15)	89.7	52.8	0.56	24.3

The direct-fusion control has no linker pLDDT because it contains zero linker residues.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 16

Slide 16 from the unchanged LinkerForge validation presentation

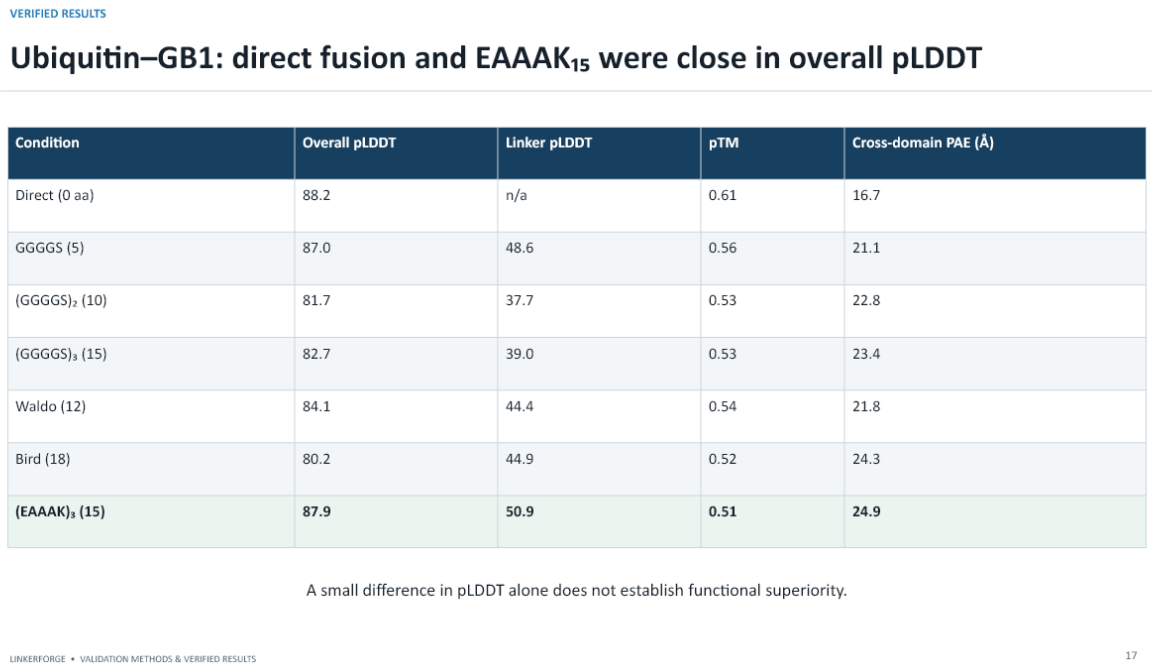
Purpose: Use this pair to show that adding a linker can reduce predictor confidence.

What to understand: Direct fusion: pLDDT 94.9 and PAE 14.4 A.

- Short GGGGS led linked constructs at pLDDT 92.2.
- All linked conditions had higher cross-domain PAE than direct fusion.

How to present it: This is the clearest evidence against a universal best-linker narrative.

Slide 17 Ubiquitin-GB1 table



VERIFIED RESULTS

Ubiquitin–GB1: direct fusion and $EAAAK_{15}$ were close in overall pLDDT

Condition	Overall pLDDT	Linker pLDDT	pTM	Cross-domain PAE (Å)
Direct (0 aa)	88.2	n/a	0.61	16.7
GGGGS (5)	87.0	48.6	0.56	21.1
$(GGGGS)_2$ (10)	81.7	37.7	0.53	22.8
$(GGGGS)_3$ (15)	82.7	39.0	0.53	23.4
Waldo (12)	84.1	44.4	0.54	21.8
Bird (18)	80.2	44.9	0.52	24.3
$(EAAAK)_3$ (15)	**87.9**	**50.9**	**0.51**	**24.9**

A small difference in pLDDT alone does not establish functional superiority.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 17

Slide 17 from the unchanged LinkerForge validation presentation

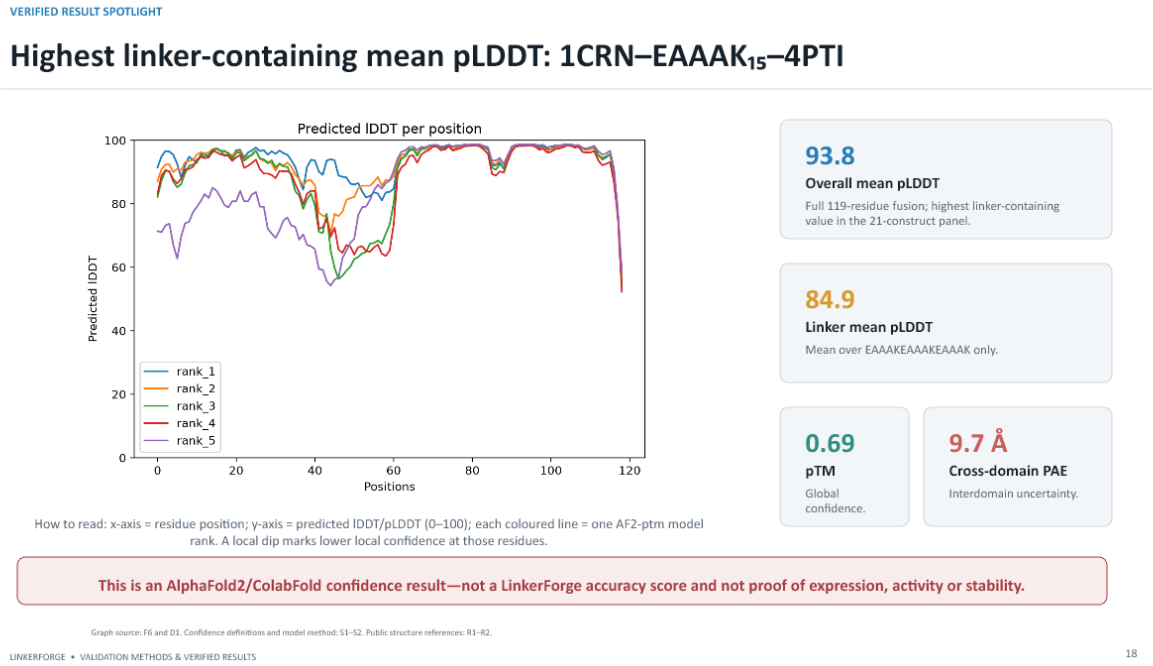
Purpose: Show that one metric can look close while another remains different.

What to understand: Direct fusion pLDDT was 88.2 with PAE 16.7 A.

- EAAAK pLDDT was 87.9 but PAE 24.9 A.
- The 0.3 pLDDT difference does not make the full confidence profile equivalent.

How to present it: Interpret several metrics jointly rather than ranking by one decimal value.

Slide 18 Highest linker-containing confidence spotlight



Slide 18 from the unchanged LinkerForge validation presentation

Purpose: Explain exactly what was highest and what the plotted lines represent.

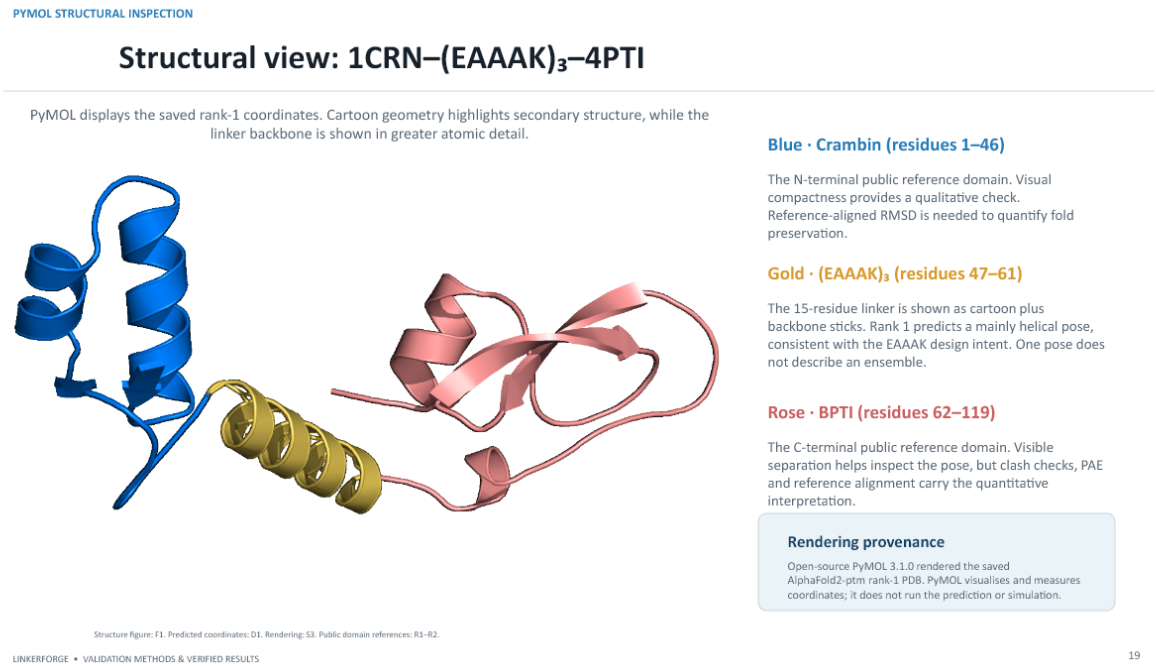
What to understand: The full fusion has 119 residues.

- Overall pLDDT 93.8; linker pLDDT 84.9; pTM 0.69; cross-domain PAE 9.7 A.
- Each coloured line is a ranked AF2-pTM output across residue positions.
- Model disagreement or local dips indicate uncertainty not visible in one average.

How to present it: State that selection was post hoc and descriptive.

Do not overclaim: Highest linker-containing mean pLDDT does not mean universally best linker.

Slide 19 PyMOL structural inspection



Slide 19 from the unchanged LinkerForge validation presentation

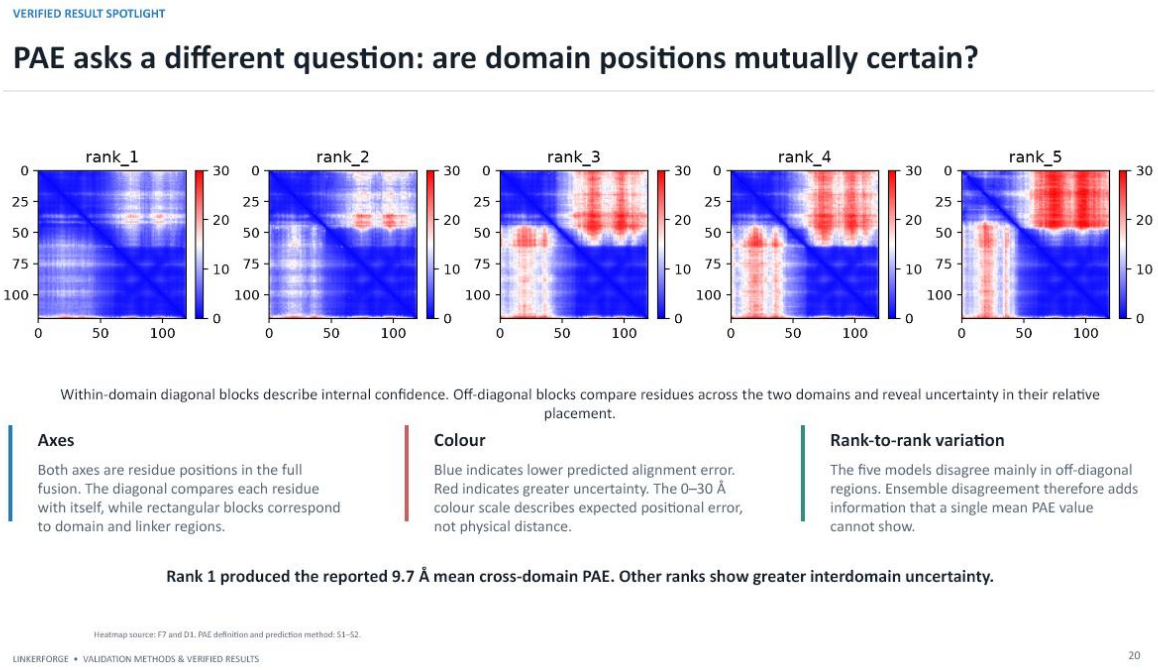
Purpose: Connect sequence ranges to the displayed three-dimensional pose.

What to understand: Crambin is residues 1-46, EAAAK15 is 47-61 and BPTI is 62-119.

- The mainly helical linker pose is consistent with the EAAAK design intent.
- One predicted pose does not represent the full conformational ensemble.
- Quantitative claims require PAE, RMSD and geometry analysis, not visual impression alone.

How to present it: Use PyMOL as an inspection and figure tool, never as a simulation engine.

Slide 20 PAE heatmaps



Slide 20 from the unchanged LinkerForge validation presentation

Purpose: Teach how the diagonal and off-diagonal blocks answer different questions.

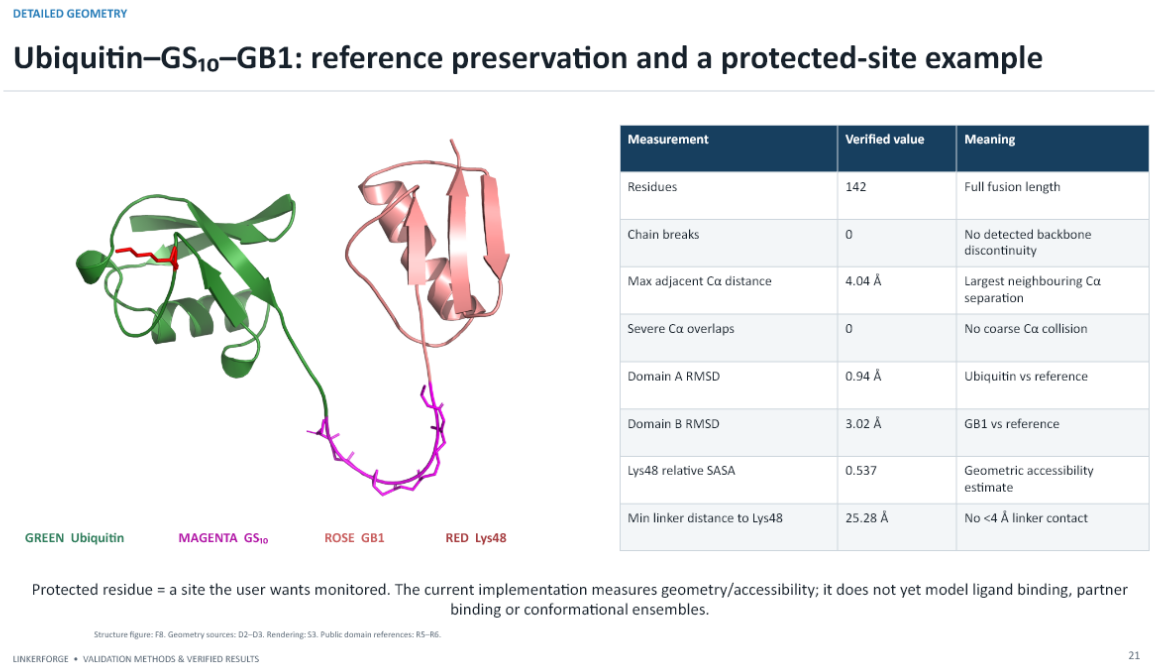
What to understand: Both axes are residue positions.

- Blue indicates lower predicted alignment error; red indicates higher uncertainty.
- Diagonal blocks describe internal regions; off-diagonal blocks describe their relative placement.
- The five ranks disagree mainly in off-diagonal regions.

How to present it: A PAE value is expected error, not physical inter-residue distance.

Do not overclaim: PAE is predicted error, not the physical distance between residues.

Slide 21 Protected-site example



Slide 21 from the unchanged LinkerForge validation presentation

Purpose: Explain how the system monitors a user-selected functional site geometrically.

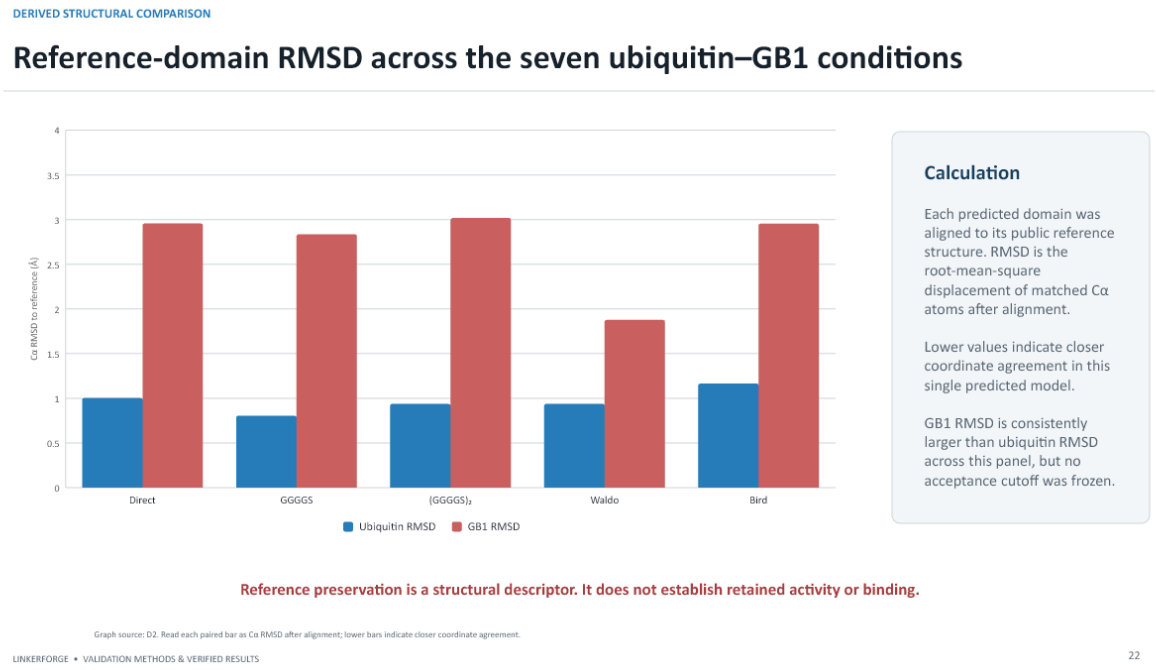
What to understand: Ubiquitin-GS10-GB1 has 142 residues and zero detected chain breaks.

- Ubiquitin RMSD was 0.94 A; GB1 RMSD was 3.02 A.
- Lys48 relative SASA was 0.537 and the nearest linker distance was 25.28 A.
- No linker contact was closer than 4 A.

How to present it: This supports geometric separation in one pose, not retained ubiquitin function.

Do not overclaim: One protected-site pose cannot prove accessibility throughout a dynamic ensemble.

Slide 22 Reference-domain RMSD



Slide 22 from the unchanged LinkerForge validation presentation

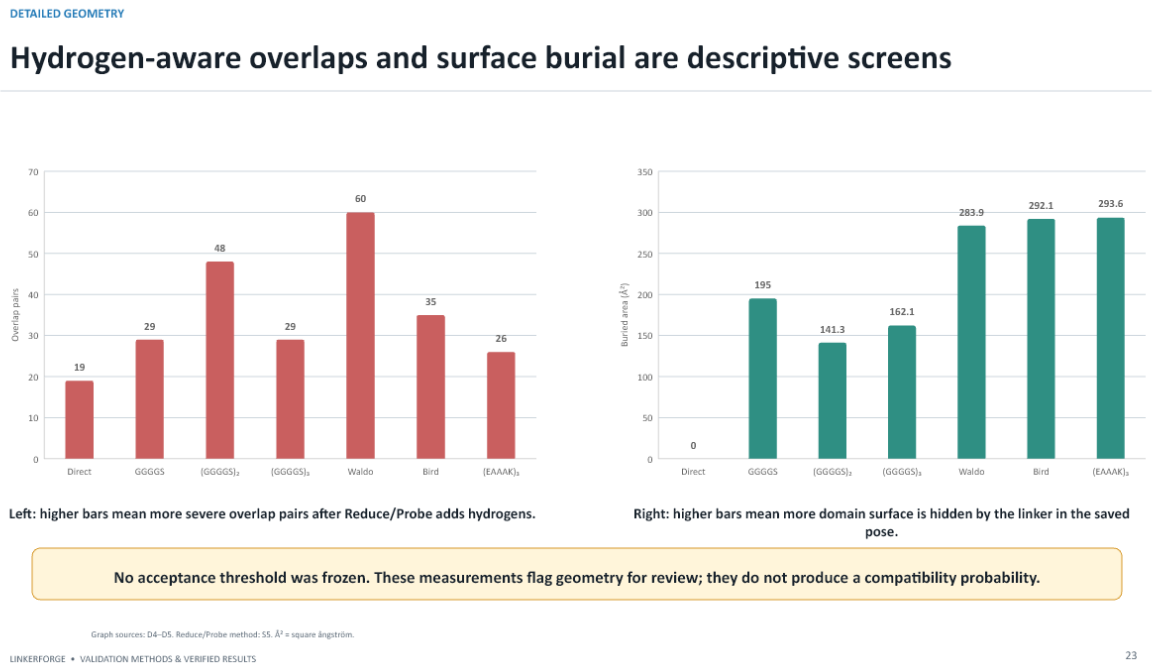
Purpose: Explain separate-domain alignment and why lower RMSD only means closer coordinates.

What to understand: Each predicted domain was aligned to its own public reference.

- Separate alignment prevents domain orientation from dominating fold-preservation measurement.
- GB1 had larger RMSD than ubiquitin across the conditions.
- No validated acceptance cutoff was frozen.

How to present it: Do not describe higher RMSD as unfolding without additional evidence.

Slide 23 Hydrogen-aware overlaps and burial



Slide 23 from the unchanged LinkerForge validation presentation

Purpose: Explain these as flags for manual review.

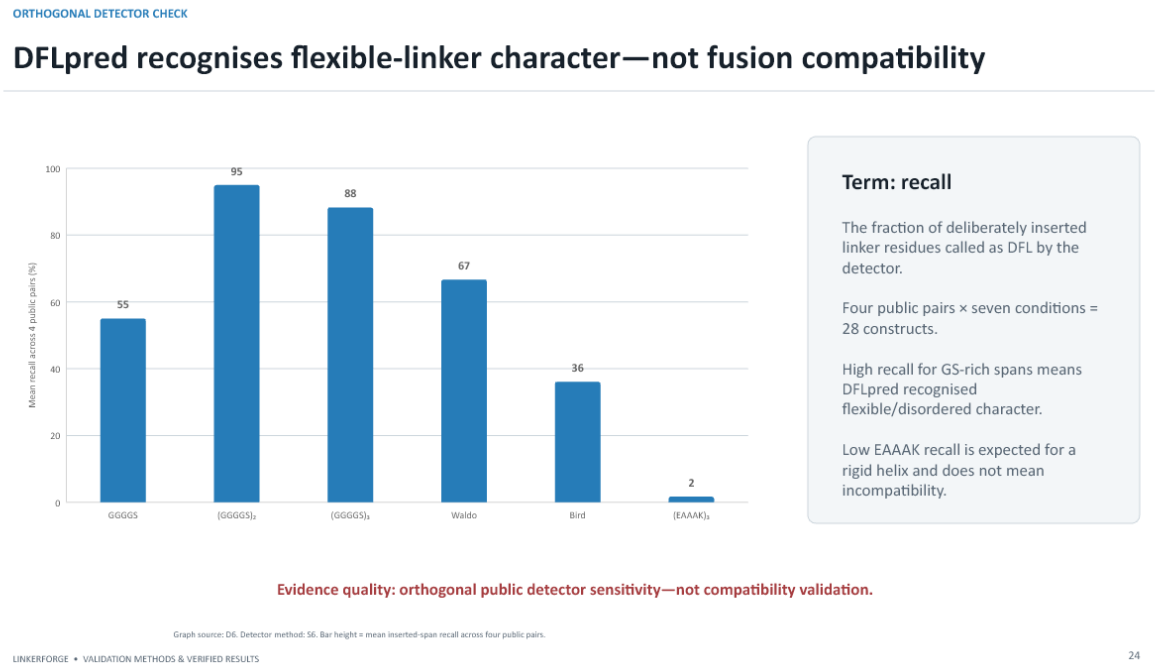
What to understand: Reduce adds or optimizes hydrogens; Probe counts unfavorable contacts.

- Waldo had the largest reported severe-overlap count in this panel.
- Bird and EAAAK had comparatively high linker-related burial.
- No frozen threshold converts these values into pass or fail.

How to present it: Packing may be favorable or obstructive depending on the intended function.

Do not overclaim: No validated overlap or burial cutoff was frozen.

Slide 24 DFLpred orthogonal check



Slide 24 from the unchanged LinkerForge validation presentation

Purpose: Explain the difference between detecting flexible-linker character and validating fusion compatibility.

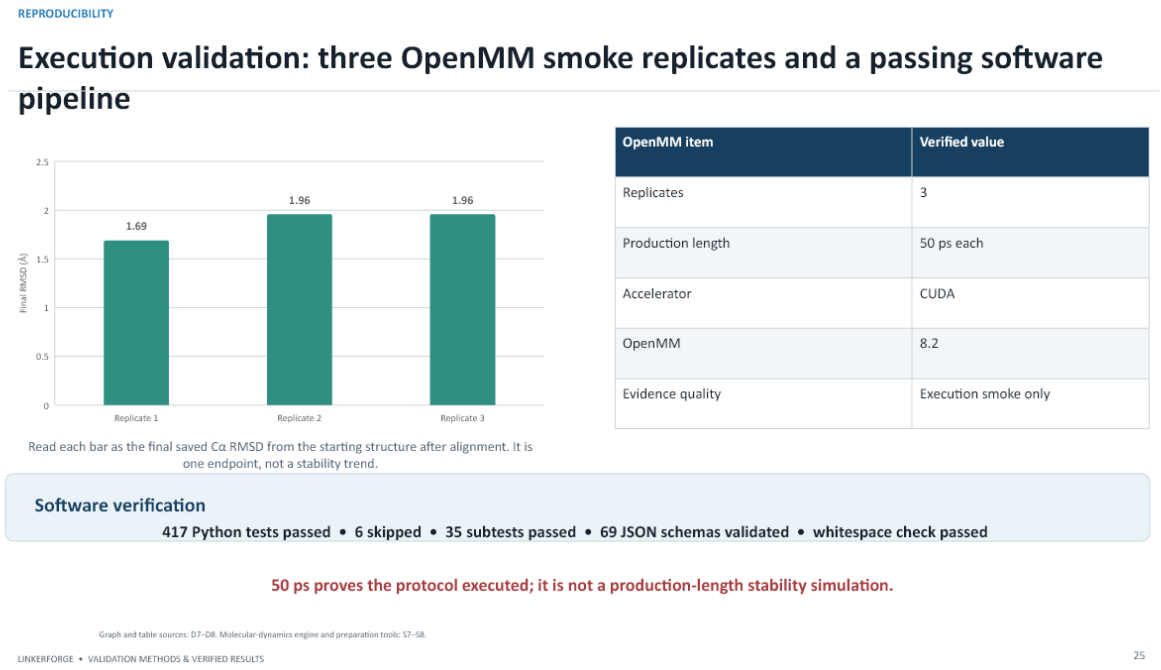
What to understand: Recall is the fraction of deliberately inserted linker residues called DFL.

- GS10 recall was 95% and GS15 88.3%.
- EAAAK recall was 1.7%, expected for a rigid helix.
- The analysis covered 4 public pairs x 7 conditions, or 28 constructs.

How to present it: Low DFL recall is not a failure for a deliberately structured linker.

Do not overclaim: DFL detection is not compatibility prediction.

Slide 25 OpenMM and software verification



Slide 25 from the unchanged LinkerForge validation presentation

Purpose: Present the simulation as an execution test and the automated tests as software evidence.

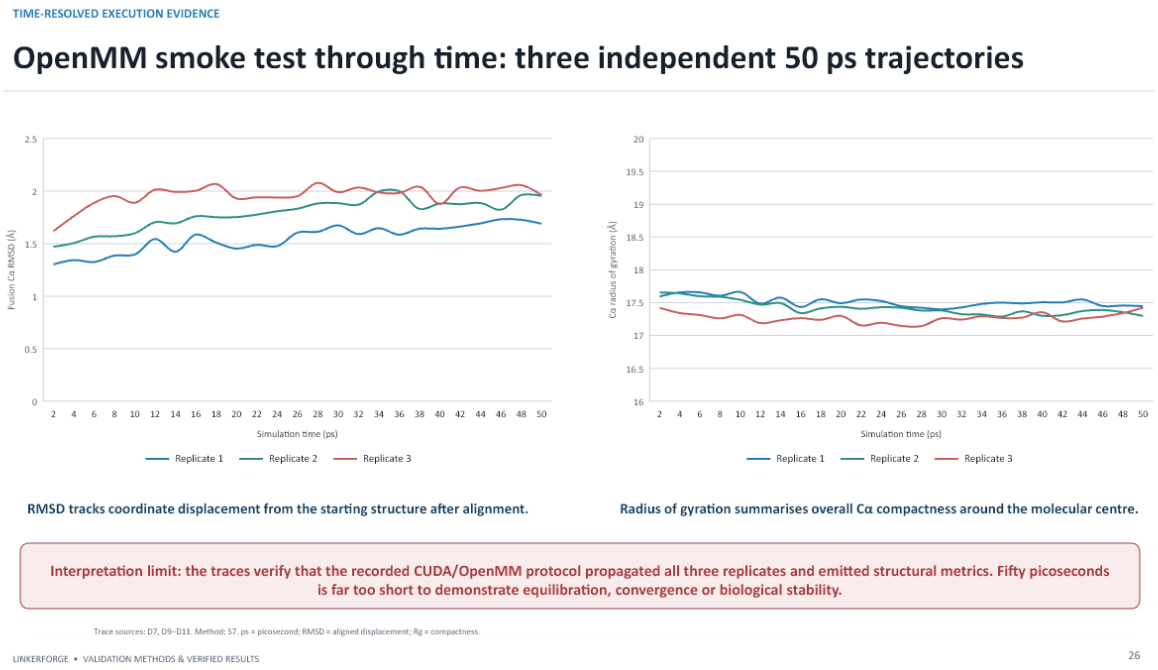
What to understand: Three independent 50 ps CUDA/OpenMM runs completed.

- Final C-alpha RMSD values were 1.69, 1.96 and 1.96 A.
- 417 Python tests passed, 6 skipped, 35 subtests passed and 69 JSON schemas validated.
- An endpoint is not a stability trend.

How to present it: Make the 50 ps limitation explicit before anyone asks.

Do not overclaim: A 50 ps smoke test is not production MD.

Slide 26 Time-resolved OpenMM traces



Slide 26 from the unchanged LinkerForge validation presentation

Purpose: Explain RMSD and radius of gyration through time.

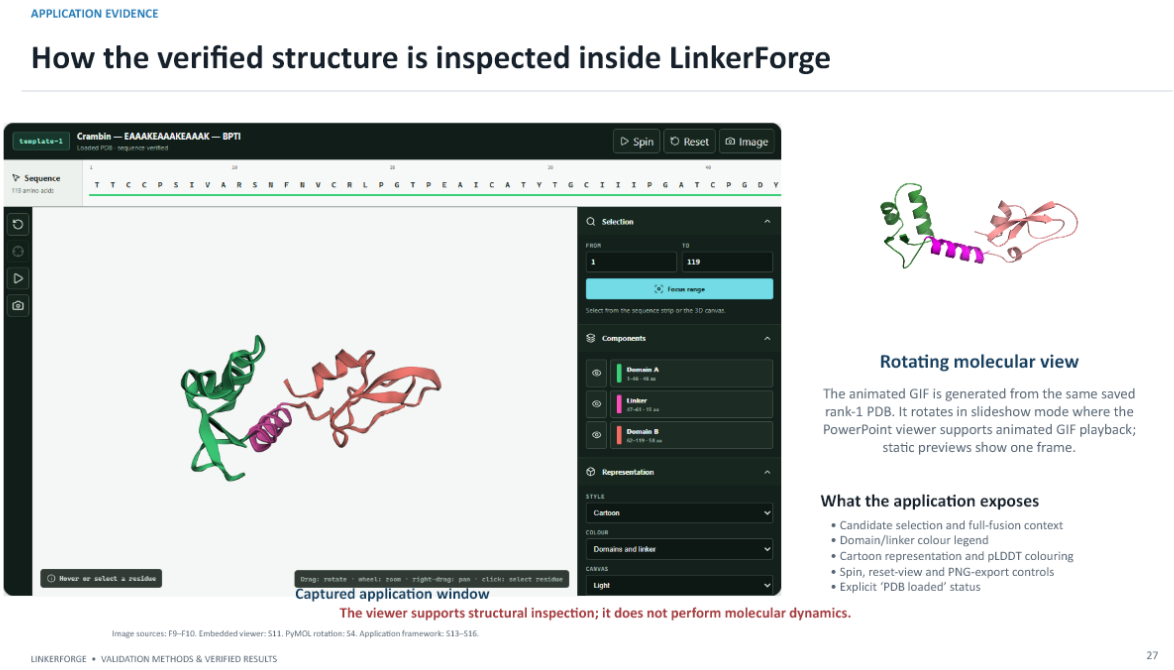
What to understand: RMSD measures aligned displacement from the starting structure.

- Rg measures overall compactness around the molecular centre.
- The traces prove that the protocol propagated coordinates and emitted metrics.
- Fifty picoseconds is too short for equilibration, convergence or biological stability.

How to present it: Production MD requires longer replicated runs and convergence analysis.

Do not overclaim: Short traces cannot demonstrate equilibrium or convergence.

Slide 27 Application evidence



Slide 27 from the unchanged LinkerForge validation presentation

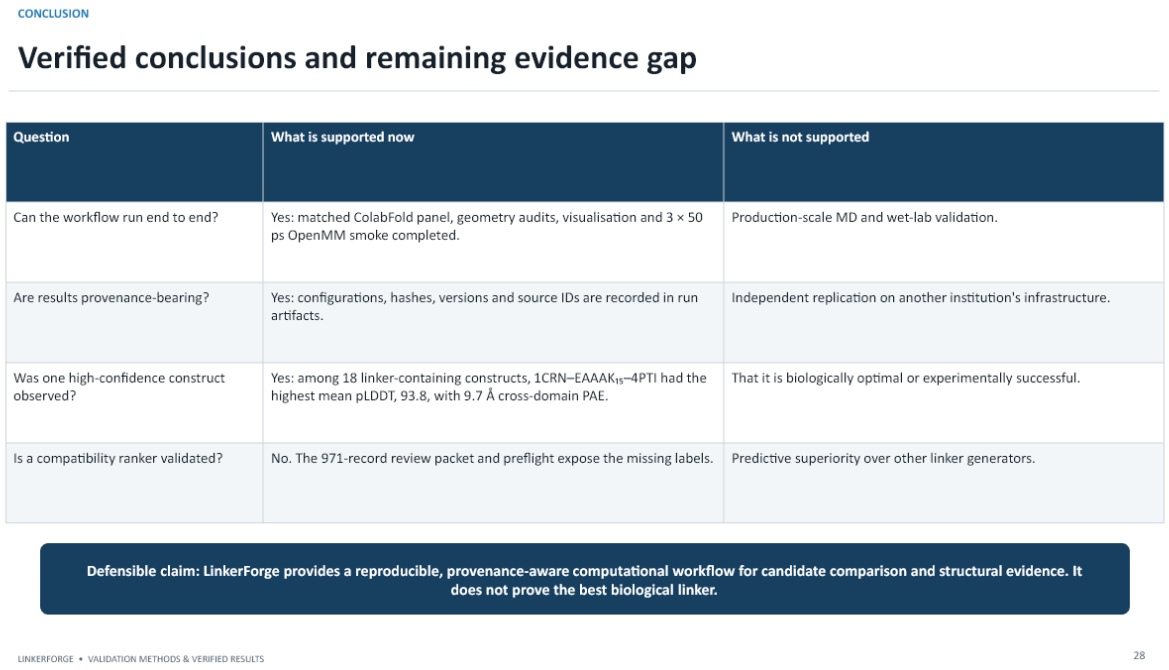
Purpose: Show how predictions and context are exposed to a researcher.

What to understand: The window combines candidate context, domain/linker colour legend and structural viewer.

- The viewer supports cartoon and pLDDT colouring, spin, reset and PNG export.
- The loaded-status message confirms that a PDB was parsed.
- The GIF and embedded viewer use saved coordinates; neither runs MD.

How to present it: This slide demonstrates usability and traceability rather than scientific superiority.

Slide 28 Verified conclusions



CONCLUSION

Verified conclusions and remaining evidence gap

Question	What is supported now	What is not supported
Can the workflow run end to end?	Yes: matched ColabFold panel, geometry audits, visualisation and 3 × 50 ps OpenMM smoke completed.	Production-scale MD and wet-lab validation.
Are results provenance-bearing?	Yes: configurations, hashes, versions and source IDs are recorded in run artifacts.	Independent replication on another institution's infrastructure.
Was one high-confidence construct observed?	Yes: among 18 linker-containing constructs, 1CRN–EAAAK$_{15}$–4PTI had the highest mean pLDDT, 93.8, with 9.7 Å cross-domain PAE.	That it is biologically optimal or experimentally successful.
Is a compatibility ranker validated?	No. The 971-record review packet and preflight expose the missing labels.	Predictive superiority over other linker generators.

Defensible claim: LinkerForge provides a reproducible, provenance-aware computational workflow for candidate comparison and structural evidence. It does not prove the best biological linker.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 28

Slide 28 from the unchanged LinkerForge validation presentation

Purpose: Summarise the strongest defensible claims and the remaining gap.

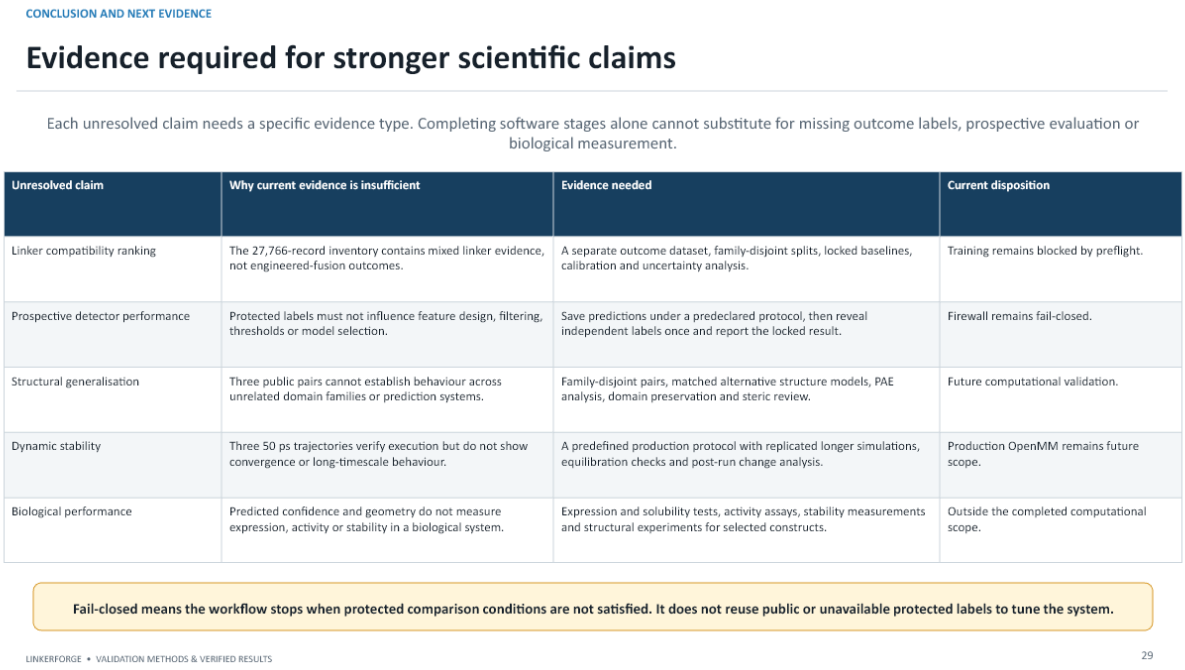
What to understand: The matched structure workflow, geometry audits, visualization and 50 ps smoke simulations executed.

- The highlighted construct is a high-confidence public prediction.
- The ranker is not scientifically trained or calibrated.
- The project does not prove a best biological linker.

How to present it: Use the exact defensible claim printed on the slide.

Do not overclaim: The conclusion must remain computational and provenance-aware.

Slide 29 Evidence needed next



CONCLUSION AND NEXT EVIDENCE

Evidence required for stronger scientific claims

Each unresolved claim needs a specific evidence type. Completing software stages alone cannot substitute for missing outcome labels, prospective evaluation or biological measurement.

Unresolved claim	Why current evidence is insufficient	Evidence needed	Current disposition
Linker compatibility ranking	The 27,766-record inventory contains mixed linker evidence, not engineered-fusion outcomes.	A separate outcome dataset, family-disjoint splits, locked baselines, calibration and uncertainty analysis.	Training remains blocked by preflight.
Prospective detector performance	Protected labels must not influence feature design, filtering, thresholds or model selection.	Save predictions under a predeclared protocol, then reveal independent labels once and report the locked result.	Firewall remains fail-closed.
Structural generalisation	Three public pairs cannot establish behaviour across unrelated domain families or prediction systems.	Family-disjoint pairs, matched alternative structure models, PAE analysis, domain preservation and steric review.	Future computational validation.
Dynamic stability	Three 50 ps trajectories verify execution but do not show convergence or long-timescale behaviour.	A predefined production protocol with replicated longer simulations, equilibration checks and post-run change analysis.	Production OpenMM remains future scope.
Biological performance	Predicted confidence and geometry do not measure expression, activity or stability in a biological system.	Expression and solubility tests, activity assays, stability measurements and structural experiments for selected constructs.	Outside the completed computational scope.

Fail-closed means the workflow stops when protected comparison conditions are not satisfied. It does not reuse public or unavailable protected labels to tune the system.

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Slide 29 from the unchanged LinkerForge validation presentation

Purpose: Map every unsupported claim to the missing evidence rather than treating all future work as one item.

What to understand: Ranker claims need engineered-fusion outcomes, family-disjoint splits, baselines and calibration.

- Prospective claims need frozen predictions before label reveal.
- Structural generalisation needs more families and model families.
- Dynamic stability needs production MD.
- Biological performance needs appropriate laboratory measurement.

How to present it: Software completion cannot substitute for outcome evidence.

Do not overclaim: Missing biological evidence cannot be replaced by additional software features.

Slide 30 How to read the figures

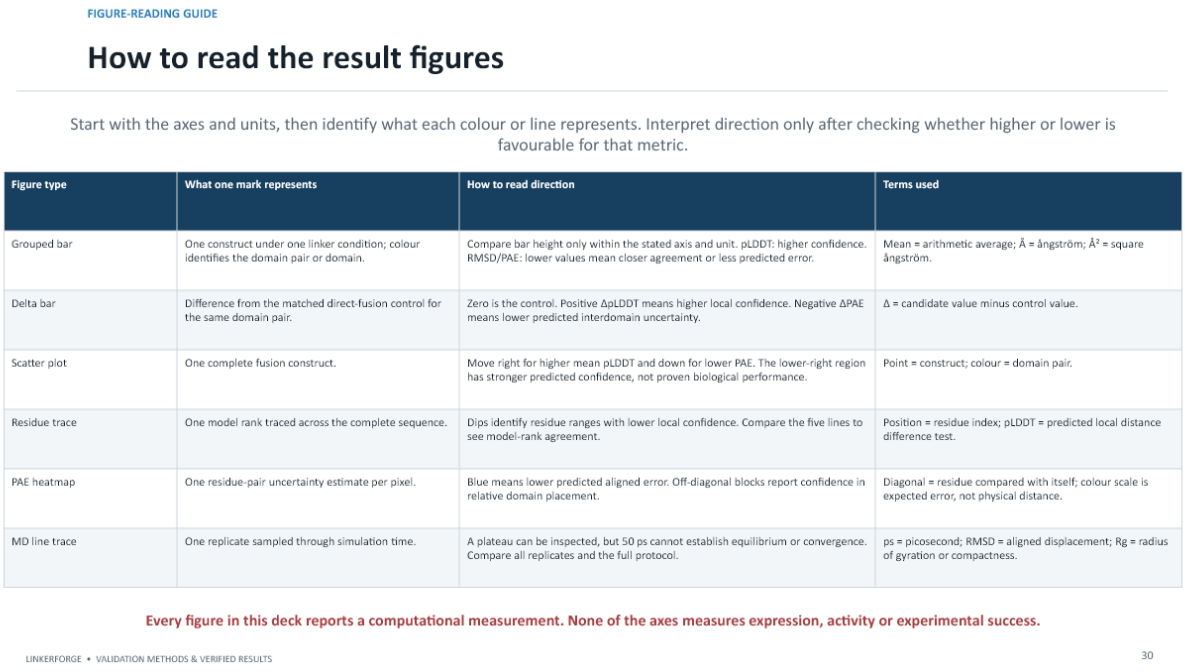


FIGURE-READING GUIDE

How to read the result figures

Start with the axes and units, then identify what each colour or line represents. Interpret direction only after checking whether higher or lower is favourable for that metric.

Figure type	What one mark represents	How to read direction	Terms used
Grouped bar	One construct under one linker condition; colour identifies the domain pair or domain.	Compare bar height only within the stated axis and unit. pLDDT: higher confidence. RMSD/PAE: lower values mean closer agreement or less predicted error.	Mean = arithmetic average; Å = ångström; Å² = square ångström.
Delta bar	Difference from the matched direct-fusion control for the same domain pair.	Zero is the control. Positive ΔpLDDT means higher local confidence. Negative ΔPAE means lower predicted interdomain uncertainty.	Δ = candidate value minus control value.
Scatter plot	One complete fusion construct.	Move right for higher mean pLDDT and down for lower PAE. The lower-right region has stronger predicted confidence, not proven biological performance.	Point = construct; colour = domain pair.
Residue trace	One model rank traced across the complete sequence.	Dips identify residue ranges with lower local confidence. Compare the five lines to see model-rank agreement.	Position = residue index; pLDDT = predicted local distance difference test.
PAE heatmap	One residue-pair uncertainty estimate per pixel.	Blue means lower predicted aligned error. Off-diagonal blocks report confidence in relative domain placement.	Diagonal = residue compared with itself; colour scale is expected error, not physical distance.
MD line trace	One replicate sampled through simulation time.	A plateau can be inspected, but 50 ps cannot establish equilibrium or convergence. Compare all replicates and the full protocol.	ps = picosecond; RMSD = aligned displacement; Rg = radius of gyration or compactness.

Every figure in this deck reports a computational measurement. None of the axes measures expression, activity or experimental success.

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Slide 30 from the unchanged LinkerForge validation presentation

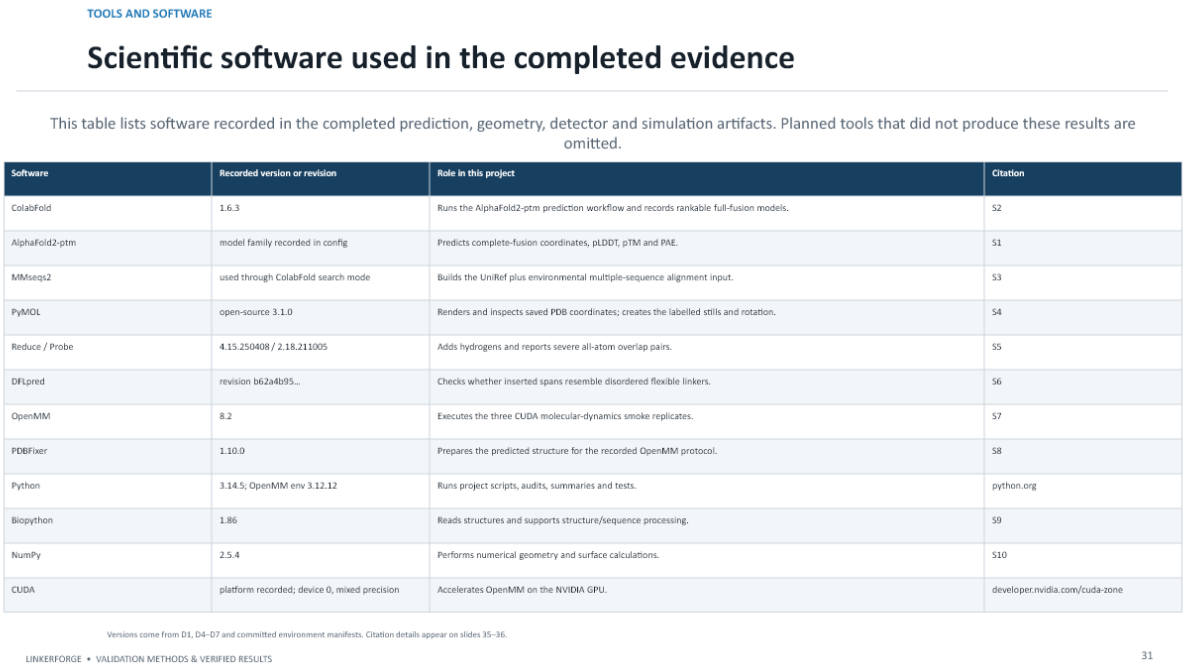
Purpose: Give the audience a reusable graph-reading method.

What to understand: Read axes and units first.

- Identify what one bar, point, line or heatmap cell represents.
- Check whether higher or lower is favorable for that metric.
- Only then interpret biological relevance, which remains limited for these computational measurements.

How to present it: None of the axes measures expression, activity or experimental success.

Slide 31 Scientific software



TOOLS AND SOFTWARE

Scientific software used in the completed evidence

This table lists software recorded in the completed prediction, geometry, detector and simulation artifacts. Planned tools that did not produce these results are omitted.

Software	Recorded version or revision	Role in this project	Citation
ColabFold	1.6.3	Runs the AlphaFold2-ptm prediction workflow and records rankable full-fusion models.	S2
AlphaFold2-ptm	model family recorded in config	Predicts complete-fusion coordinates, pLDDT, pTM and PAE.	S1
MMseqs2	used through ColabFold search mode	Builds the UniRef plus environmental multiple-sequence alignment input.	S3
PyMOL	open-source 3.1.0	Renders and inspects saved PDB coordinates; creates the labelled stills and rotation.	S4
Reduce / Probe	4.15.250408 / 2.18.211005	Adds hydrogens and reports severe all-atom overlap pairs.	S5
DFLpred	revision b62a4b95...	Checks whether inserted spans resemble disordered flexible linkers.	S6
OpenMM	8.2	Executes the three CUDA molecular-dynamics smoke replicates.	S7
PDBFixer	1.10.0	Prepares the predicted structure for the recorded OpenMM protocol.	S8
Python	3.14.5; OpenMM env 3.12.12	Runs project scripts, audits, summaries and tests.	python.org
Biopython	1.86	Reads structures and supports structure/sequence processing.	S9
NumPy	2.5.4	Performs numerical geometry and surface calculations.	S10
CUDA	platform recorded; device 0, mixed precision	Accelerates OpenMM on the NVIDIA GPU.	developer.nvidia.com/cuda-zone

Versions come from D1, D4–D7 and committed environment manifests. Citation details appear on slides 35–36.

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Slide 31 from the unchanged LinkerForge validation presentation

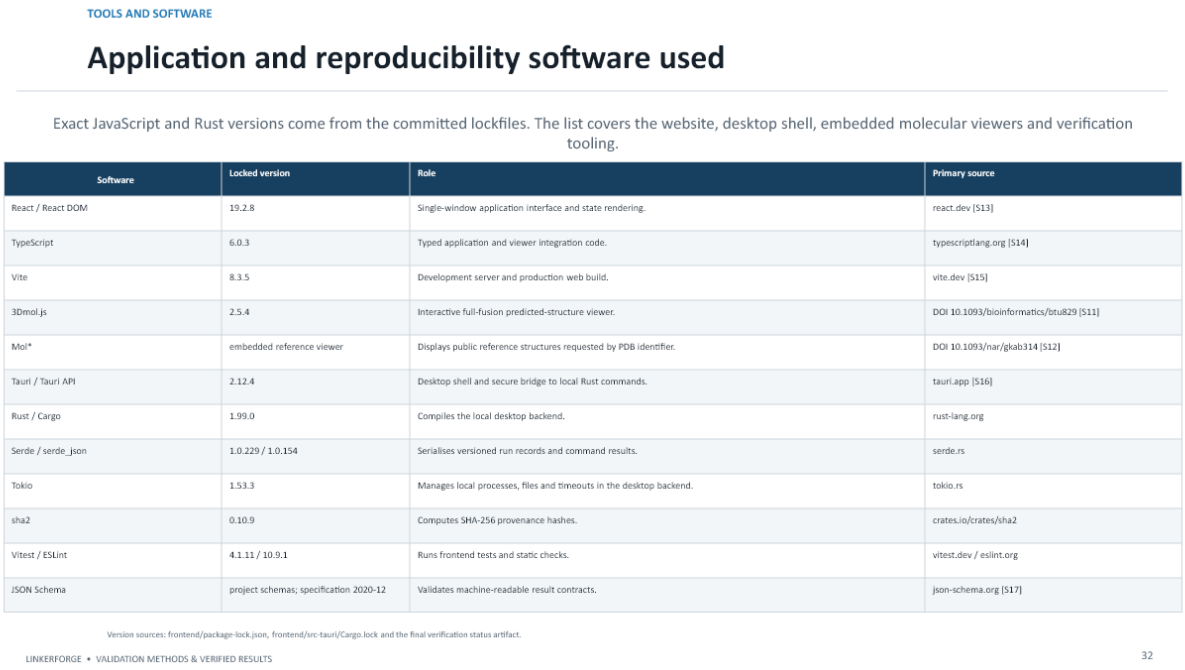
Purpose: Explain the scientific role of each tool and why its version is recorded.

What to understand: ColabFold and AF2-pTM predict structures and confidence.

- MMseqs2 creates sequence alignments.
- PyMOL visualises coordinates.
- Reduce/Probe inspect atomic contacts.
- DFLpred tests flexible-linker character.
- OpenMM and PDBFixer prepare and execute the MD smoke workflow.

How to present it: Versions make reruns interpretable because algorithms and defaults can change.

Slide 32 Application and reproducibility software



TOOLS AND SOFTWARE

Application and reproducibility software used

Exact JavaScript and Rust versions come from the committed lockfiles. The list covers the website, desktop shell, embedded molecular viewers and verification tooling.

Software	Locked version	Role	Primary source
React / React DOM	19.2.8	Single-window application interface and state rendering.	react.dev [513]
TypeScript	6.0.3	Typed application and viewer integration code.	typescriptlang.org [514]
Vite	8.3.5	Development server and production web build.	vite.dev [515]
3Dmol.js	2.5.4	Interactive full-fusion predicted-structure viewer.	DOI 10.1093/bioinformatics/btu829 [511]
Mol*	embedded reference viewer	Displays public reference structures requested by PDB identifier.	DOI 10.1093/nar/gkab314 [512]
Tauri / Tauri API	2.12.4	Desktop shell and secure bridge to local Rust commands.	tauri.app [516]
Rust / Cargo	1.99.0	Compiles the local desktop backend.	rust-lang.org
Serde / serde_json	1.0.229 / 1.0.154	Serialises versioned run records and command results.	serde.rs
Tokio	1.53.3	Manages local processes, files and timeouts in the desktop backend.	tokio.rs
sha2	0.10.9	Computes SHA-256 provenance hashes.	crates.io/crates/sha2
Vitest / ESLint	4.1.11 / 10.9.1	Runs frontend tests and static checks.	vitest.dev / eslint.org
JSON Schema	project schemas; specification 2020-12	Validates machine-readable result contracts.	json-schema.org [517]

Version sources: frontend/package-lock.json, frontend/src-tauri/Cargo.lock and the final verification status artifact.

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Slide 32 from the unchanged LinkerForge validation presentation

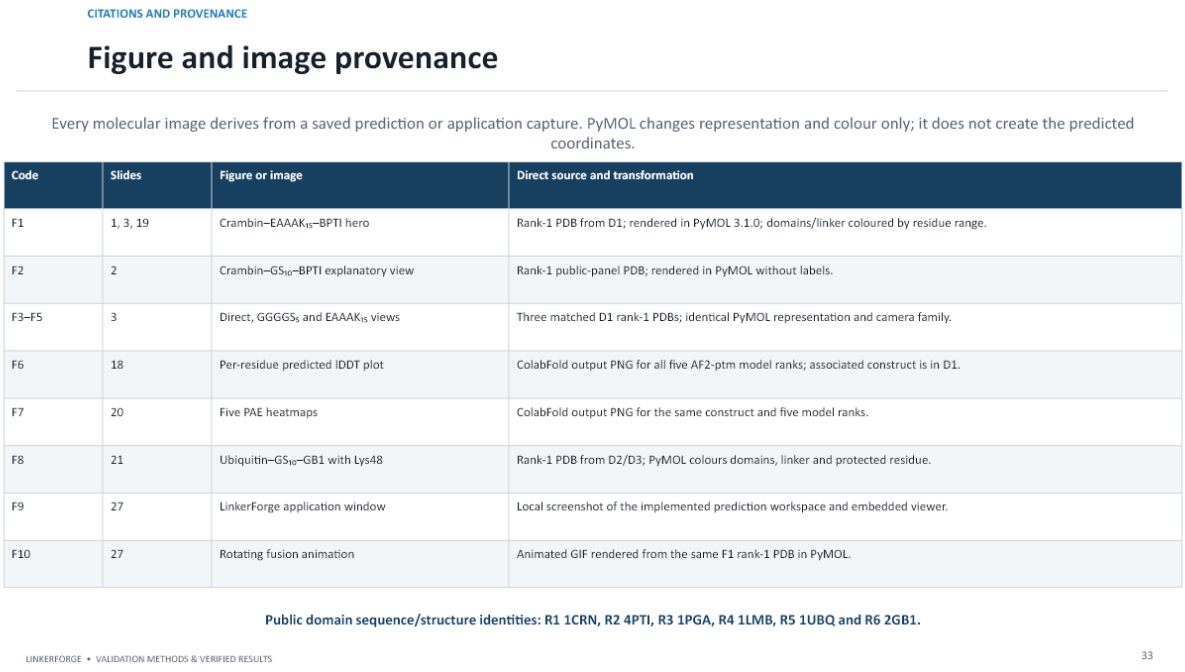
Purpose: Explain how the user interface and native desktop functions are constructed.

What to understand: React and TypeScript implement the interface; Vite builds it.

- 3Dmol.js and Mol* provide embedded molecular viewing.
- Tauri packages the interface as a desktop application with Rust-native commands.
- Serde, Tokio and sha2 support structured data, asynchronous work and hashes.
- Vitest, ESLint and JSON Schema support verification.

How to present it: These tools provide the application, not the biological evidence.

Slide 33 Figure and image provenance



CITATIONS AND PROVENANCE

Figure and image provenance

Every molecular image derives from a saved prediction or application capture. PyMOL changes representation and colour only; it does not create the predicted coordinates.

Code	Slides	Figure or image	Direct source and transformation
F1	1, 3, 19	Crambin–EAAAK$_{15}$–BPTI hero	Rank-1 PDB from D1; rendered in PyMOL 3.1.0; domains/linker coloured by residue range.
F2	2	Crambin–GS$_{10}$–BPTI explanatory view	Rank-1 public-panel PDB; rendered in PyMOL without labels.
F3–F5	3	Direct, GGGGS$_5$ and EAAAK$_{15}$ views	Three matched D1 rank-1 PDBs; identical PyMOL representation and camera family.
F6	18	Per-residue predicted lDDT plot	ColabFold output PNG for all five AF2-ptm model ranks; associated construct is in D1.
F7	20	Five PAE heatmaps	ColabFold output PNG for the same construct and five model ranks.
F8	21	Ubiquitin–GS$_{10}$–GB1 with Lys48	Rank-1 PDB from D2/D3; PyMOL colours domains, linker and protected residue.
F9	27	LinkerForge application window	Local screenshot of the implemented prediction workspace and embedded viewer.
F10	27	Rotating fusion animation	Animated GIF rendered from the same F1 rank-1 PDB in PyMOL.

Public domain sequence/structure identities: R1 1CRN, R2 4PTI, R3 1PGA, R4 1LMB, R5 1UBQ and R6 2GB1.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS

33

Slide 33 from the unchanged LinkerForge validation presentation

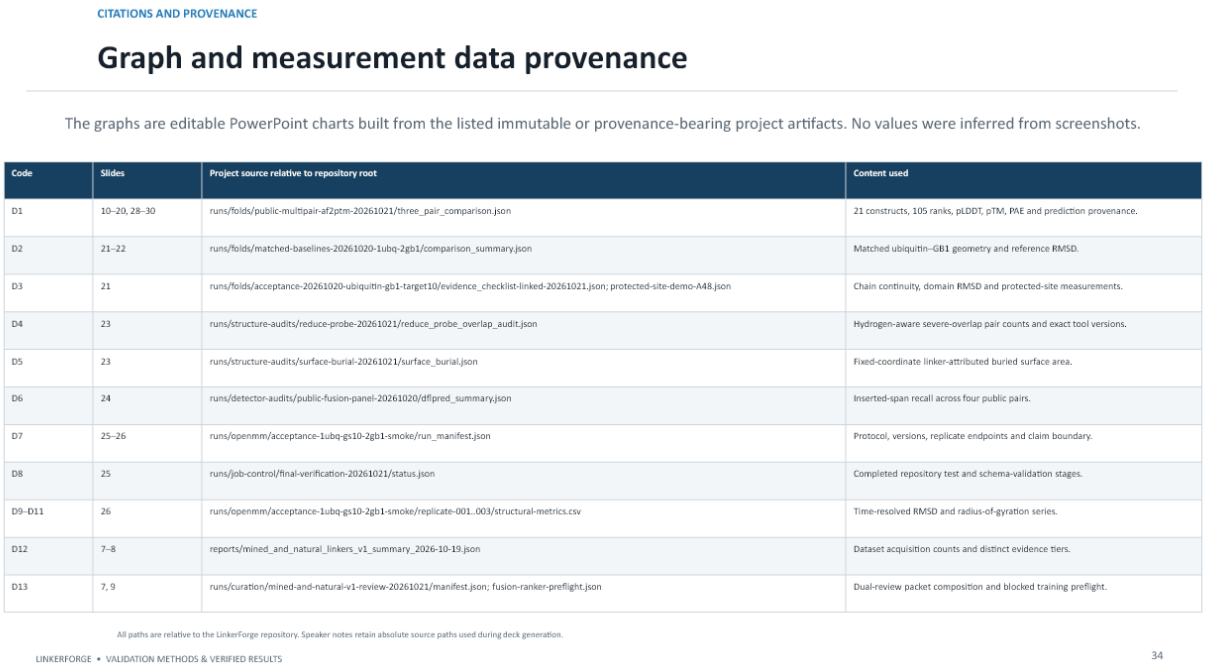
Purpose: Explain that every image is traceable to saved coordinates or an application capture.

What to understand: F codes identify figures and images.

- Public identities are tied to RCSB PDB entries.
- PyMOL changed representation and colour only.
- The figure register allows an examiner to trace a visual to its source file and transformation.

How to present it: Provenance supports auditability; it does not guarantee biological truth.

Slide 34 Graph and measurement provenance



CITATIONS AND PROVENANCE

Graph and measurement data provenance

The graphs are editable PowerPoint charts built from the listed immutable or provenance-bearing project artifacts. No values were inferred from screenshots.

Code	Slides	Project source relative to repository root	Content used
D1	10–20, 28–30	runs/folds/public-multipair-af2ptm-20261021/three_pair_comparison.json	21 constructs, 105 ranks, pLDDT, pTM, PAE and prediction provenance.
D2	21–22	runs/folds/matched-baselines-20261020-1ubq-2gb1/comparison_summary.json	Matched ubiquitin–GB1 geometry and reference RMSD.
D3	21	runs/folds/acceptance-20261020-ubiquitin-gb1-target10/evidence_checklist-linked-20261021.json; protected-site-demo-A48.json	Chain continuity, domain RMSD and protected-site measurements.
D4	23	runs/structure-audits/reduce-probe-20261021/reduce_probe_overlap_audit.json	Hydrogen-aware severe-overlap pair counts and exact tool versions.
D5	23	runs/structure-audits/surface-burial-20261021/surface_burial.json	Fixed-coordinate linker-attributed buried surface area.
D6	24	runs/detector-audits/public-fusion-panel-20261020/dflpred_summary.json	Inserted-span recall across four public pairs.
D7	25–26	runs/openmm/acceptance-1ubq-gs10-2gb1-smoke/run_manifest.json	Protocol, versions, replicate endpoints and claim boundary.
D8	25	runs/job-control/final-verification-20261021/status.json	Completed repository test and schema-validation stages.
D9–D11	26	runs/openmm/acceptance-1ubq-gs10-2gb1-smoke/replicate-001..003/structural-metrics.csv	Time-resolved RMSD and radius-of-gyration series.
D12	7–8	reports/mined_and_natural_linkers_v1_summary_2026-10-19.json	Dataset acquisition counts and distinct evidence tiers.
D13	7, 9	runs/curation/mined-and-natural-v1-review-20261021/manifest.json; fusion-ranker-preflight.json	Dual-review packet composition and blocked training preflight.

All paths are relative to the LinkerForge repository. Speaker notes retain absolute source paths used during deck generation.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 34

Slide 34 from the unchanged LinkerForge validation presentation

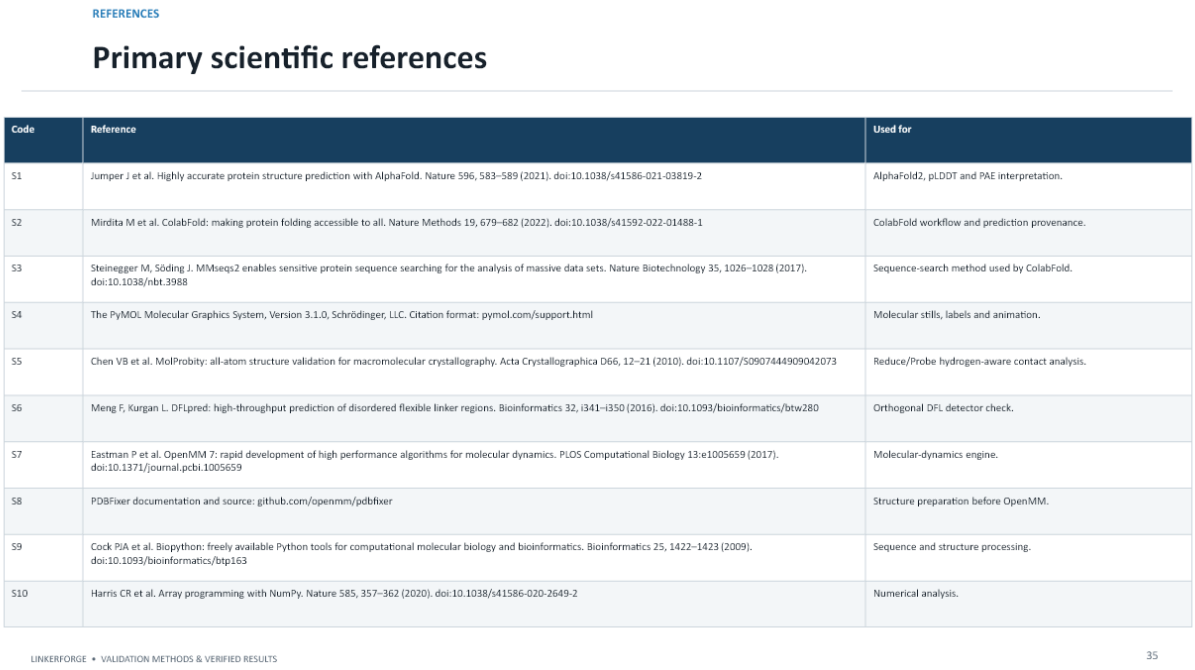
Purpose: Explain that chart values came from saved project artifacts rather than manual reading of pictures.

What to understand: D codes identify data artifacts.

- The PowerPoint charts are editable but sourced from persisted values.
- Speaker notes retain the absolute paths used during deck generation.
- Hashes and versioned schemas protect result identity.

How to present it: This is how the project separates presentation from the underlying evidence.

Slide 35 Primary scientific references



REFERENCES

Primary scientific references

Code	Reference	Used for
S1	Jumper J et al. Highly accurate protein structure prediction with AlphaFold. Nature 596, 583–589 (2021). doi:10.1038/s41586-021-03819-2	AlphaFold2, pLDDT and PAE interpretation.
S2	Mirdita M et al. ColabFold: making protein folding accessible to all. Nature Methods 19, 679–682 (2022). doi:10.1038/s41592-022-01488-1	ColabFold workflow and prediction provenance.
S3	Steinegger M, Söding J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. Nature Biotechnology 35, 1026–1028 (2017). doi:10.1038/nbt.3988	Sequence-search method used by ColabFold.
S4	The PyMOL Molecular Graphics System, Version 3.1.0, Schrödinger, LLC. Citation format: pymol.com/support.html	Molecular stills, labels and animation.
S5	Chen VB et al. MolProbity: all-atom structure validation for macromolecular crystallography. Acta Crystallographica D66, 12–21 (2010). doi:10.1107/S0907444909042073	Reduce/Probe hydrogen-aware contact analysis.
S6	Meng F, Kurgan L. DFLpred: high-throughput prediction of disordered flexible linker regions. Bioinformatics 32, i341–i350 (2016). doi:10.1093/bioinformatics/btw280	Orthogonal DFL detector check.
S7	Eastman P et al. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. PLOS Computational Biology 13:e1005659 (2017). doi:10.1371/journal.pcbi.1005659	Molecular-dynamics engine.
S8	PDBFixer documentation and source: github.com/openmm/pdbfixer	Structure preparation before OpenMM.
S9	Cock PJA et al. Biopython: freely available Python tools for computational molecular biology and bioinformatics. Bioinformatics 25, 1422–1423 (2009). doi:10.1093/bioinformatics/btp163	Sequence and structure processing.
S10	Harris CR et al. Array programming with NumPy. Nature 585, 357–362 (2020). doi:10.1038/s41586-020-2649-2	Numerical analysis.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 35

Slide 35 from the unchanged LinkerForge validation presentation

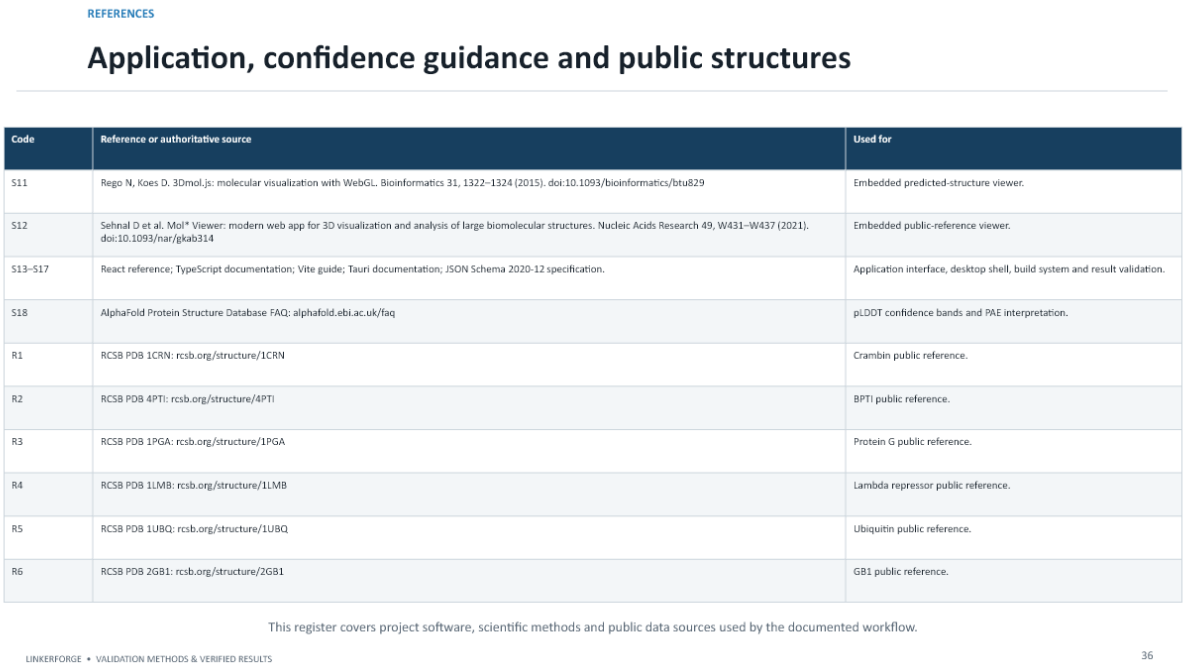
Purpose: Use this as the method register rather than reading every citation aloud.

What to understand: Primary references support AlphaFold2, ColabFold, structure assessment, DFL detection and molecular dynamics.

- When challenged on a method, identify the relevant source code and reference.
- Citation supports the method definition, while the project artifact supports the reported value.

How to present it: Distinguish literature authority from evidence produced in this project.

Slide 36 Application, confidence and public structures



REFERENCES

Application, confidence guidance and public structures

Code	Reference or authoritative source	Used for
S11	Rego N, Koes D. 3Dmol.js: molecular visualization with WebGL. Bioinformatics 31, 1322–1324 (2015). doi:10.1093/bioinformatics/btu829	Embedded predicted-structure viewer.
S12	Sehnal D et al. Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures. Nucleic Acids Research 49, W431–W437 (2021). doi:10.1093/nar/gkab314	Embedded public-reference viewer.
S13–S17	React reference; TypeScript documentation; Vite guide; Tauri documentation; JSON Schema 2020-12 specification.	Application interface, desktop shell, build system and result validation.
S18	AlphaFold Protein Structure Database FAQ: alphafold.ebi.ac.uk/faq	pLDDT confidence bands and PAE interpretation.
R1	RCSB PDB 1CRN: rcsb.org/structure/1CRN	Crambin public reference.
R2	RCSB PDB 4PTI: rcsb.org/structure/4PTI	BPTI public reference.
R3	RCSB PDB 1PGA: rcsb.org/structure/1PGA	Protein G public reference.
R4	RCSB PDB 1LMB: rcsb.org/structure/1LMB	Lambda repressor public reference.
R5	RCSB PDB 1UBQ: rcsb.org/structure/1UBQ	Ubiquitin public reference.
R6	RCSB PDB 2GB1: rcsb.org/structure/2GB1	GB1 public reference.

This register covers project software, scientific methods and public data sources used by the documented workflow.

LINKERFORGE • VALIDATION METHODS & VERIFIED RESULTS 36

Slide 36 from the unchanged LinkerForge validation presentation

Purpose: Close the traceability chain for software guidance and public protein sources.

What to understand: The register includes authoritative confidence guidance, application software and RCSB entries.

- R codes identify 1CRN, 4PTI, 1PGA, 1LMB, 1UBQ and 2GB1.
- The complete register makes the deck auditable without implying every source validates the final fusion.

How to present it: End by returning to the evidence boundary and inviting questions.

Detailed numerical reference for slides 15 to 17

Domain pair	Condition	Mean pLDDT	Linker pLDDT	pTM	Cross-domain PAE A
1CRN-4PTI	Direct	74.5	N/A	0.62	17.3
1CRN-4PTI	GGGGS	87.6	51.2	0.60	16.2
1CRN-4PTI	GS10	86.4	48.0	0.55	20.1
1CRN-4PTI	Waldo12	83.4	46.4	0.55	20.4
1CRN-4PTI	EAAAK15	93.8	84.9	0.69	9.7
1CRN-4PTI	GS15	84.2	52.3	0.50	24.1
1CRN-4PTI	Bird18	85.0	60.9	0.57	18.8
1PGA-1LMB	Direct	94.9	N/A	0.66	14.4
1PGA-1LMB	GGGGS	92.2	48.9	0.59	21.5
1PGA-1LMB	GS10	90.2	40.5	0.57	23.8
1PGA-1LMB	Waldo12	87.9	41.6	0.55	25.4
1PGA-1LMB	EAAAK15	89.7	52.8	0.56	24.3
1PGA-1LMB	GS15	88.6	36.9	0.54	26.0
1PGA-1LMB	Bird18	89.7	64.9	0.54	26.0
1UBQ-2GB1	Direct	88.2	N/A	0.61	16.7
1UBQ-2GB1	GGGGS	87.0	48.6	0.56	21.1
1UBQ-2GB1	GS10	81.7	37.7	0.53	22.8
1UBQ-2GB1	Waldo12	84.1	44.4	0.54	21.8
1UBQ-2GB1	EAAAK15	87.9	50.9	0.51	24.9
1UBQ-2GB1	GS15	82.7	39.0	0.53	23.4
1UBQ-2GB1	Bird18	80.2	44.9	0.52	24.3

Use this table for revision, not as a script to read aloud. The main pattern is that results depend on both the linker and the domain pair. Crambin-BPTI EAAAK is the strongest linker-containing confidence profile, whereas Protein G-repressor and Ubiquitin-GB1 show why direct fusion remains an essential control.

Part Three Complete presentation speech

The following speech is written for approximately 24 to 30 minutes at a clear scientific speaking pace. It is intentionally shorter than the study guide. Pause on figures, point to the relevant axis or coloured structure and do not read the reference tables aloud.

Slides 1 to 3 Introduction and biological context

Good morning. Our project is LinkerForge, a computational workflow for designing and evaluating peptide linkers used to join two protein regions into one fusion protein. This presentation reports only the computational evidence that we completed. It does not claim that a selected linker is experimentally proven or universally optimal.

A fusion protein is one continuous amino-acid chain consisting of Domain A, an optional linker and Domain B. The linker may influence separation, motion, orientation and the accessibility of important sites. On slide 2, Crambin is shown in blue, the GS10 linker in gold and BPTI in rose. The displayed structure is an AlphaFold2-family prediction rendered in PyMOL, not an experimental crystal structure.

We compare three broad design ideas: direct fusion, a flexible glycine-serine linker and a more rigid EAAAK helical linker. Direct fusion is the zero-linker baseline. GGGGS is intended to permit motion, while repeated EAAAK is intended to form a more persistent spacer. The differences shown here are descriptive results for the same Crambin-BPTI pair and are not proof that one linker class is generally superior.

Slides 4 to 6 Confidence metrics and evidence boundaries

Before showing results, I need to distinguish the confidence metrics. pLDDT is AlphaFold's per-residue local confidence on a zero-to-one-hundred scale. We report both the overall mean and the mean over linker residues. pTM estimates confidence in global topology, while PAE estimates positional uncertainty after alignment. Cross-domain PAE therefore helps us assess how consistently the two domains are arranged relative to each other. Lower PAE is preferable, but it does not prove that the orientation is biologically correct.

The geometry metrics answer different questions. C-alpha continuity screens for coordinate gaps. Reference RMSD measures coordinate deviation after alignment. SASA describes solvent exposure, burial describes surface hidden in the predicted pose and Reduce plus Probe provide a hydrogen-aware contact screen. OpenMM molecular dynamics generates coordinates through time, and radius of gyration describes overall compactness.

These evidence levels must not be confused. A technically reproducible run can still lack compatibility labels or biological confirmation. The protected benchmark therefore uses a fail-closed rule: if prospective independence is not satisfied, the workflow stops instead of using exposed labels to tune the system.

Slides 7 to 9 Dataset and curation

The dataset inventory combines several evidence categories without treating them as equivalent. We scanned 161,317 full-length proteins and obtained 27,595 mined candidate spans. We also collected 91 structure-supported CATH or PDB natural spans, 57 curated DisProt flexible-linker records and 23 published engineered templates. Together these form 27,766 records. A separate 40-protein no-linker specificity panel is not included in that total.

This is a provenance-rich linker inventory, not a fusion-compatibility ground-truth dataset. Most records tell us that a sequence may be a linker or occurs naturally between domains. They do not tell us whether inserting that linker between arbitrary engineered partners improves expression, activity or stability.

We prepared a 971-record packet for two independent curators, covering 563 family clusters and 105 sampling strata. However, the preflight correctly reports that scientific training is not authorised because outcome-linked compatibility labels and a valid family-disjoint fusion split are absent. Curation can improve linker identity and provenance; it cannot create missing experimental outcomes.

Slides 10 to 14 Matched public comparison design

For the completed public acceptance run, we selected three public domain pairs and evaluated seven conditions per pair: direct fusion, three GS lengths, two published linker sequences and the EAAAK helical control. This produced 21 complete fusion sequences. Five AlphaFold2-pTM outputs were retained for every construct, giving 105 structures. These are ranked prediction outputs, not five independent laboratory replicates.

All conditions used one fixed ColabFold protocol: ColabFold 1.6.3, AlphaFold2-pTM, MMseqs2 sequence searching, one seed, no templates and ranking by pLDDT. Therefore, the run asks how structural-confidence descriptors vary across linkers and partner pairs under matched settings.

The grouped pLDDT graph shows that the same linker does not behave identically across pairs. The delta graphs subtract each pair's direct-fusion value. Positive delta pLDDT means higher local confidence, while negative delta PAE means reduced predicted interdomain uncertainty. The scatter plot combines these two ideas: moving right means higher local confidence and moving down means lower interdomain uncertainty. Even the lower-right region represents predictor confidence, not biological performance.

Slides 15 to 18 Numerical results

For Crambin-BPTI, EAAAK15 produced the strongest linker-containing profile: overall pLDDT 93.8, linker pLDDT 84.9, pTM 0.69 and cross-domain PAE 9.7 angstroms. For Protein G-repressor, direct fusion was strongest overall at 94.9 pLDDT, while short GGGGS led the linker-containing conditions at 92.2. For Ubiquitin-GB1, direct fusion and EAAAK were close in mean pLDDT, but direct fusion had substantially lower cross-domain PAE.

These results are important because they reject a simple universal-winner interpretation. The effect of a linker depends on its partner context and on which metric is examined.

The highlighted construct is 1CRN-EAAAK15-4PTI. It was the highest mean-pLDDT linker-containing result in the 21-construct panel. The residue plot shows confidence across all 119 residues for the five ranked models. This is an AlphaFold2 confidence result, not a LinkerForge accuracy score and not proof of expression, activity or stability.

Slides 19 to 24 Structural inspection and orthogonal checks

PyMOL displays the rank-1 coordinates with Crambin in blue, EAAAK in gold and BPTI in rose. The linker appears mainly helical, which agrees with its design intent, but one predicted pose cannot describe the entire solution ensemble.

The PAE heatmaps ask whether residue positions are mutually certain. Blue diagonal blocks indicate lower internal uncertainty. Off-diagonal blocks describe relative domain placement. The model ranks disagree mainly in these off-diagonal regions, so the relative arrangement is less certain than the local domain folds.

For Ubiquitin-GS10-GB1, we also tested a protected-site workflow using Lys48. The saved model had no detected chain break, ubiquitin reference RMSD of 0.94 angstrom, GB1 RMSD of 3.02 angstrom, Lys48 relative SASA of 0.537 and a minimum linker distance of 25.28 angstrom. This supports geometric separation in one pose, not retained Lys48 function across a dynamic ensemble.

Hydrogen-aware Probe overlaps and linker-related surface burial were then used as descriptive flags. No validated threshold was frozen. Finally, DFLpred was used as an independent flexible-linker-character check. GS-rich sequences showed high recall, whereas EAAAK showed 1.7 percent recall, which is expected for a structured helical linker. DFLpred does not test fusion compatibility.

Slides 25 to 27 OpenMM and application

We ran three independent 50 picosecond OpenMM smoke simulations on CUDA. Their final C-alpha RMSD values were 1.69, 1.96 and 1.96 angstrom. The time traces show RMSD and radius of gyration, but 50 picoseconds is far too

short to establish equilibration, convergence or long-term stability. These runs verify that the preparation, simulation and metric-extraction workflow executes.

The software pipeline also passed 417 Python tests, with 6 skipped, 35 subtests passed, 69 JSON schemas validated and the whitespace check passed.

The application slide shows how a researcher can inspect a selected full-fusion model without leaving LinkerForge. It exposes the candidate context, domain and linker colours, cartoon and pLDDT representations, rotation, view reset, PNG export and explicit PDB-loaded status. The viewer displays saved coordinates; it does not run molecular dynamics.

Slides 28 to 36 Conclusions, future evidence and provenance

The completed work supports an end-to-end computational evidence workflow: matched structure prediction, confidence extraction, visual inspection, geometry measurements, an orthogonal detector check, short OpenMM execution tests and provenance-bearing records. The strongest highlighted linker-containing prediction was Crambin-EAAAK15-BPTI.

It does not support a claim that LinkerForge predicts the biologically best linker. A scientifically trained ranker still requires engineered-fusion outcome data, family-disjoint evaluation, matched baselines, calibration and prospective testing. Structural generalisation requires more protein families and additional model families. Dynamic-stability claims require production-length replicated molecular dynamics. Biological performance requires the appropriate experiments.

The remaining slides explain how to read each graph, identify all scientific and application software and trace every image and numerical result to a saved source. The reference register separates the literature supporting each method from the project artifacts supporting each reported value.

In conclusion, LinkerForge currently provides a reproducible and provenance-aware platform for linker-candidate comparison and structural-evidence review. It integrates more of the practical workflow in one application, while keeping the claim boundary explicit. Thank you, and I welcome your questions.

Part Four Difficult viva and examiner questions

Answer the direct question first. Then explain the evidence and finish with the limitation. The longer answers below are for understanding; during the presentation, use the first two or three sentences unless the examiner asks for more detail.

What is the scientific novelty of LinkerForge?

The defensible novelty is workflow integration and evidence discipline. LinkerForge keeps partner-specific inputs, candidate and control construction, matched structure prediction, confidence interpretation, structural inspection, geometry checks and provenance in one application. Predictive superiority is not yet established.

What is the LinkerForge model architecture?

The frozen proposed ranker design uses frozen ESM-2 protein embeddings, attachment-point geometric features, a small linker encoder and multitask ranking heads. Its learning task is pairwise or contrastive compatibility ranking. The presentation does not claim that this ranker has been scientifically trained or calibrated.

Is LinkerForge a generative AI system?

The present MVP creates candidates by constrained enumeration, mutation and retrieval. Learned sequence generation and diffusion are future work. It is therefore inaccurate to describe the completed evidence as validation of a learned generator.

Did LinkerForge achieve 93.8 percent accuracy?

No. 93.8 is the mean AlphaFold2 pLDDT of one construct. It is local structural confidence averaged across residues, not classification accuracy, compatibility probability or an experimental success rate.

Why is EAAAK described as the best result?

It is not described as universally best. It was the highest mean-pLDDT linker-containing construct in the completed public panel and had a favorable PAE for Crambin-BPTI. Other pairs did not show the same pattern.

Why did you exclude the direct-fusion row from the highlighted result?

The highlight explicitly answers a linker-design question, so it selects among constructs containing a linker. The direct-fusion control is still reported and produced the highest overall pLDDT row for Protein G-repressor.

Why is direct fusion necessary?

It provides the zero-linker baseline. Without it, an apparent improvement or deterioration could not be interpreted relative to simply joining the two domains.

Why can low linker pLDDT be acceptable?

A flexible linker may occupy many conformations rather than one stable local structure. AlphaFold can therefore be uncertain. Low pLDDT can be consistent with intended flexibility, although it does not prove flexibility or function.

How can pLDDT be high while PAE is also high?

The two domains can each be locally well folded, producing high pLDDT, while their relative orientation remains uncertain, producing high cross-domain PAE.

Is PAE the distance between two residues?

No. PAE is expected positional error after aligning on another residue. It uses angstrom units but does not report physical separation.

Why use pTM as well as PAE?

pTM compresses global topology confidence into one value, while the PAE matrix reveals where the uncertainty occurs. Cross-domain PAE specifically focuses on relative placement of the partners.

Why were only three domain pairs used?

They were deliberately short public examples used to verify the complete workflow under available compute. They are an acceptance panel, not a representative estimate of performance across protein space.

Are five AlphaFold outputs five replicates?

No. They are related ranked outputs produced by AlphaFold2 model parameterisations under the same input. They reveal some model disagreement but are not independent biological or statistical replicates.

Why use one seed?

One seed made the matched public acceptance run computationally feasible. It limits the uncertainty analysis; a stronger benchmark would include multiple seeds and predeclared aggregation rules.

Why disable structural templates?

Using no templates kept the protocol uniform and reduced the risk that one construct had a favorable template unavailable to the others. It does not make the prediction unbiased; all outputs still come from the same model family.

Why is the 27,766-record dataset not enough?

Most records label possible or natural linker identity, not the outcome of an engineered fusion. Compatibility training requires partner-linker-partner constructs linked to measured outcomes.

What would a proper compatibility dataset contain?

It would contain both partner sequences, attachment positions, exact linker sequence, experimental conditions and measured outcomes such as expression, solubility, activity, stability or structural integrity. It should also record negative and inconclusive outcomes rather than only successful publications.

Why are natural linkers not sufficient ground truth?

Natural multidomain proteins evolved in their native context. An engineered fusion may combine unrelated surfaces, termini and functions. Natural linkers are valuable references and possible pretraining data but are not direct labels for arbitrary engineered compatibility.

What do the curators validate?

They review whether a proposed span is a defensible linker and whether its source and evidence labels are correct. They do not determine whether it will succeed between every pair of engineered proteins.

Why do you need two curators?

Independent review reduces individual bias and enables agreement measurement. Adjudication then resolves disagreements transparently.

What is a family-disjoint split?

Closely related protein families are kept on only one side of the train-test boundary. This prevents near-duplicate biology from making test performance look better than true generalisation.

What is calibration?

Calibration compares predicted probability with observed frequency. A well-calibrated 0.8 prediction should succeed about 80 percent of the time in comparable cases. The current workflow does not have calibrated compatibility probabilities.

Why did the protected benchmark fail closed?

The conditions for an independent prospective test were not satisfied. The workflow therefore refused to use exposed or unavailable protected labels to guide model selection. This preserves the future validity of the benchmark.

Why not use public benchmark scores to choose the model?

Once a public or protected result influences model design, threshold selection or feature choice, it becomes development feedback and no longer provides an unbiased final evaluation.

Does DFLpred prove that the linkers are correct?

It provides an orthogonal check of flexible or disordered linker character. It does not measure fusion compatibility, domain preservation or biological function.

Why did EAAAK fail DFLpred?

It did not fail its design goal. DFLpred looks for disordered flexible-linker character, while EAAAK is intended to form a helix. Low DFL recall is therefore expected.

What does zero chain breaks mean?

The coordinate file did not contain a gap exceeding the implemented C-alpha continuity rule. It supports coordinate continuity but does not validate every bond angle, side chain or energy term.

Does low RMSD prove that the domain is functional?

No. RMSD measures coordinate similarity after alignment. Function can depend on a small number of residues, dynamics, ligands or interactions that a global C-alpha RMSD does not capture.

Why was GB1 RMSD larger?

The predicted GB1 coordinates deviated more from the selected public reference under the defined alignment. The calculation does not establish whether the cause is fusion context, model uncertainty, reference differences or another factor.

Does the Lys48 analysis prove accessibility?

It shows that Lys48 remained geometrically separated from the linker and partially solvent accessible in one saved pose. It does not prove accessibility across a conformational ensemble or during partner binding.

Are more Probe overlaps always worse?

They indicate more severe close-contact pairs after hydrogen placement and deserve review. They are not a validated universal rejection score, and relaxation may change some contacts.

Is greater surface burial always undesirable?

No. Burial may obstruct a functional surface or create favorable packing. It must be interpreted using the intended biological function and protected sites.

Why use OpenMM for only 50 ps?

The objective was execution validation: preparation, CUDA dynamics, replication and metric output. The result does not support a stability claim.

What would production MD require?

Longer simulations, multiple replicas, explicit equilibration criteria, convergence analysis, suitable force-field and solvent choices, trajectory quality control and predeclared post-run metrics. The required duration depends on the question and system.

Why use PyMOL if the application has a viewer?

The embedded viewer supports convenient user inspection. PyMOL provides reproducible selections, measurements and high-quality scientific renders. Neither tool performs AlphaFold prediction or molecular dynamics.

Can AlphaFold predict linker dynamics?

AlphaFold predicts coordinates and confidence, not a time-resolved ensemble. Low confidence or model disagreement may indicate uncertainty, but it is not a molecular-dynamics trajectory.

Why was the highlighted construct selected after the run?

It was chosen post hoc to demonstrate the application using the strongest linker-containing mean-pLDDT result. The deck labels that selection as descriptive and reports the complete panel, so it is not presented as an independent winning benchmark.

Is LinkerForge better than existing linker tools?

The present evidence supports broader workflow integration and provenance. It does not establish predictive superiority because no locked family-disjoint outcome benchmark has compared all methods under identical rules.

Could the method favour structured linkers because AlphaFold is more confident about helices?

Yes, that is a plausible model-related confound. AlphaFold confidence may favor regions with stable predicted structure, whereas a functional flexible linker may appropriately remain uncertain. This is why biological outcomes and matched linker classes are needed.

Why not rank only by mean pLDDT?

Mean pLDDT can hide a weak linker, ignore interdomain uncertainty and overlook steric or functional-site problems. A defensible funnel must consider several metrics and ultimately outcome evidence.

What is the strongest claim you can make today?

LinkerForge executes a reproducible, provenance-aware workflow for matched candidate comparison and structural-evidence review, and it produced a clearly documented high-confidence public example. It has not yet proven the biologically best linker.

What is required for publication?

A software or methods paper may be possible with accurate scoping, reproducible artifacts and transparent limitations. A predictive-superiority paper requires outcome-linked data, family-disjoint prospective evaluation, calibration, matched external baselines and appropriate biological validation.

Part Five Rapid glossary

Term	Meaning in this project
Adjudication	Resolution of disagreements after independent curators complete their reviews.
Amino acid	A chemical building block of a protein.
AlphaFold2-pTM	An AlphaFold2 configuration that predicts coordinates and global-topology confidence.
Angstrom A	A distance unit equal to 10^-10 metres.
Baseline	A predefined reference condition used for comparison.
Calibration	Agreement between predicted probabilities and observed frequencies.
Candidate	One sequence or design being evaluated.
C-alpha	The central backbone carbon used as one reference point per amino-acid residue.
ColabFold	A practical AlphaFold-family workflow using fast MMseqs2 sequence searching.
Compatibility ranker	A model intended to order linkers for a particular pair of fusion partners.
Condition	One exact linker choice applied to one domain pair.
Construct	The complete Domain A-linker-Domain B amino-acid sequence.
Control	A comparison chosen before interpretation, such as direct fusion or a standard GS repeat.
Cross-domain PAE	Mean predicted alignment uncertainty between the two domain regions.
CUDA	NVIDIA GPU-computing platform used by the OpenMM smoke runs.
Curator	A subject-matter reviewer who examines the biological identity and supporting evidence of a record.
DFL	Disordered flexible linker.
Domain	A protein region expected to adopt a recognisable fold.
Evidence quality	A label describing source support and curation status, not probability of biological success.
Family-disjoint	Related protein families are prevented from appearing on both sides of a train-test split.
Fail closed	Stop when protected conditions are not satisfied rather than weaken the scientific rule.
Fusion protein	One amino-acid chain created by joining two protein regions.
Hash	A file fingerprint used to detect changes and identify exact artifacts.

Term	Meaning in this project
Linker	A covalently encoded peptide placed between fusion partners.
MMseqs2	Fast protein-sequence search and alignment software.
MSA	Multiple sequence alignment containing evolutionary sequence relationships.
OpenMM	A molecular-dynamics simulation engine.
PAE	Predicted aligned error, an AlphaFold estimate of relative positional uncertainty.
PDB	Protein Data Bank or the coordinate-file format commonly used for macromolecular structures.
pLDDT	AlphaFold per-residue local confidence on a 0-100 scale.
Prospective	Predictions and rules are frozen before the protected outcome is revealed.
Protected residue	A site selected for monitoring because accessibility or geometry may be functionally important.
Provenance	The source, processing, configuration and version history of a result.
pTM	AlphaFold predicted global-topology confidence on a 0-1 scale.
Radius of gyration	A measure of overall atomic compactness around the molecular centre.
Recall	The fraction of true target residues or cases that a detector identifies.
Replicate	A separately executed run under a defined protocol.
Residue	An amino acid incorporated into a protein chain.
RMSD	Root-mean-square coordinate deviation after structural alignment.
SASA	Solvent-accessible surface area.
Schema	A machine-readable definition of required result fields and formats.
Smoke test	A short execution test that validates the workflow rather than the full scientific hypothesis.
Steric overlap	Atoms occupying physically incompatible space in a structural model.

Final claim boundary checklist

Avoid: Our model achieved 93.8 percent accuracy.

Use: The construct had a mean AlphaFold2 pLDDT of 93.8.

Avoid: EAAAK is the best linker.

Use: EAAAK was the highest mean-pLDDT linker-containing condition in this public panel.

Avoid: The 27,766 records are experimentally validated compatibility data.

Use: They are a mixed-evidence, provenance-labelled linker inventory.

Avoid: DFLpred confirmed compatibility.

Use: DFLpred detected flexible-linker character in inserted spans.

Avoid: OpenMM proved stability.

Use: Three 50 ps OpenMM smoke replicates validated execution.

Avoid: PyMOL simulated the protein.

Use: PyMOL rendered and inspected saved coordinates.

Avoid: The protected benchmark failed because the model was bad.

Use: The benchmark failed closed because independence requirements were not satisfied.

Avoid: LinkerForge outperforms current methods.

Use: LinkerForge integrates a broader provenance-aware workflow; superiority remains to be tested.

Thirty second closing summary

LinkerForge has demonstrated a reproducible workflow for constructing and comparing fusion-linker candidates, running matched AlphaFold2/ColabFold predictions, interpreting multiple confidence measures, inspecting structures, measuring geometry, checking flexible-linker character and validating the OpenMM execution path. The strongest linker-containing public prediction was 1CRN-EAAAK15-4PTI, but this remains computational evidence. A trained and calibrated compatibility ranker, prospective family-disjoint evaluation, production dynamics and biological outcomes are still required for stronger claims.